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[Intervention Review]

(Ultra-)short-acting insulin analogues for adults with type 1 diabetes mellitus on multiple daily injections: a network meta-analysis

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ABSTRACT

Rationale

Ultra-short-acting insulin analogues have the potential to achieve clinical benefits for people with type 1 diabetes, but their relative effects in relation to regular human insulin remain unclear.

Objectives

Primary objective: To compare the effects of long-term treatment with (ultra-)short-acting insulin analogues to regular human insulin or another (ultra-)short-acting insulin analogue in people with type 1 diabetes on multiple daily injections through a network meta-analysis.

Secondary objective: To obtain an estimate of the relative ranking of these (ultra-)short-acting insulin analogues compared to regular human insulin.

Search methods

We searched CENTRAL, MEDLINE, Web of Science, the WHO International Clinical Trials Registry Platform and ClinicalTrials.gov from inception to 14 May 2025, without language restrictions.

Eligibility criteria

We included all randomised controlled trials (RCTs) with an intervention duration of at least 24 weeks that compared (ultra-)short-acting insulin analogues with regular human insulins in non-pregnant adults with type 1 diabetes on multiple daily injections.

Outcomes

Critical outcomes included glycaemic control, hypoglycaemia, diabetic ketoacidosis, quality of life and adherence to treatment.

Risk of bias

We evaluated studies with the Cochrane risk of bias tool (RoB 2).

Synthesis methods

We performed statistical analyses using a frequentist network meta-analysis to compare estimates for the critical outcomes, with regular human insulin as the reference treatment, and a random-effects model for other outcomes and pairwise comparisons. We rated the certainty of the evidence using the CINeMA framework.

Included studies

We included 15 RCTs with 6335 participants evaluating insulin aspart, lispro, glulisine, ultra-rapid lispro (URLi) and fast-acting insulin aspart (FIAsp). The duration of interventions ranged from 24 to 52 weeks. Participants had a mean age of 36.4 years and a mean HbA1c of 8.1%. Participants had different basal insulin regimens. Thirteen studies were funded by the pharmaceutical industry.

Synthesis of results

Network meta-analysis: (Ultra-)short-acting insulin analogues versus regular human insulin (RHI)

Glycaemic control: haemoglobin A1c (HbA1c)

Based on low-certainty evidence from 14 studies with 6039 participants, all five (ultra-)short-acting insulin analogues may result in little to no difference in HbA1c compared to RHI (mean difference (MD): URLi -0.25%, 95% confidence interval (CI) -0.51% to 0.00%; lispro -0.22%, 95% CI -0.46% to 0.02%; glulisine -0.21%, 95% CI -0.48% to 0.06%; FIAsp -0.17%, 95% CI -0.28% to -0.05%; aspart -0.14%, 95% CI -0.21% to -0.06%).

Glycaemic control: fasting blood glucose (FBG)

Based on extremely low-certainty evidence from four studies with 2737 participants, we are uncertain about the effects of aspart and FIAsp compared to RHI (MD: FIAsp 0.69 mmol/L, 95% CI -0.31 mmol/L to 1.69 mmol/L; aspart 0.80 mmol/L, 95% CI -0.01 mmol/L to 1.61 mmol/L). Lispro, glulisine and URLi could not be assessed in relation to RHI.

Mild/moderate (non-severe) hypoglycaemia

Based on very low-certainty evidence from six studies with 3967 participants, we are uncertain about the effects of aspart and FIAsp compared to RHI (risk ratio (RR): aspart 1.05, 95% CI 0.97 to 1.13; FIAsp 1.04, 95% CI 0.96 to 1.12). Lispro, glulisine and URLi could not be assessed in relation to RHI.

Severe/serious hypoglycaemia

Based on low- to very low-certainty evidence from 11 studies with 5390 participants, URLi may result in a reduction in severe/serious hypoglycaemia compared to RHI (RR: URLi 0.80, 95% CI 0.36 to 1.77), but the evidence is very uncertain for aspart and FIAsp (RR: aspart 0.89, 95% CI 0.73 to 1.09; FIAsp 0.79, 95% CI 0.55 to 1.15). Glulisine may result in an increase in severe/serious hypoglycaemia compared to RHI (RR glulisine 1.54, 95% CI 0.46 to 5.19). In contrast, lispro may result in little to no difference compared to RHI (RR: lispro 1.01, 95% CI 0.51 to 1.98).

Mild/moderate nocturnal hypoglycaemia

This outcome could not be evaluated with network meta-analysis.

Severe/serious nocturnal hypoglycaemia

Based on very low-certainty evidence from four studies with 3229 participants, we are uncertain about the effects of aspart and FIAsp compared to RHI (RR: aspart 0.63, 95% CI 0.46 to 0.88; FIAsp 0.52, 95% CI 0.26 to 1.04). Lispro, glulisine and URLi could not be assessed in relation to RHI.

Diabetic ketoacidosis (DKA)

Based on low- to very low-certainty evidence from nine studies with 4599 participants, lispro and URLi may result in a reduction in DKA compared to RHI (odds ratio (OR): lispro 0.74, 95% CI 0.10 to 5.39; URLi 0.74, 95% CI 0.03 to 16.64), whereas FIAsp may result in an increase compared to RHI (OR: FIAsp 1.18, 95% CI 0.12 to 11.24), but the evidence is very uncertain for aspart (OR: aspart 0.80, 95% CI 0.20 to 3.11). Glulisine could not be assessed in relation to RHI.

Quality of life

This outcome could not be evaluated with network meta-analysis.

Adherence to treatment

Based on very low-certainty evidence from five studies with 3579 participants, we are uncertain about the effects of aspart and FIAsp compared to RHI (RR: aspart 1.01, 95% CI 0.95 to 1.07; FIAsp 1.01, 95% CI 0.94 to 1.08). Lispro, glulisine and URLi could not be assessed in relation to RHI.

Authors' conclusions

(Ultra-)short-acting insulin analogues may result in little to no difference in glycaemic control, non-severe hypoglycaemia and treatment adherence at short-term follow-up, but some agents may reduce severe/serious hypoglycaemia and DKA. We are uncertain about the effects of these insulins on mild to moderate nocturnal hypoglycaemia and quality of life. There were no long-term follow-up data.

Funding

World Health Organization (WHO).

Registration

Protocol for this update: PROSPERO CRD42024599817 (<https://www.crd.york.ac.uk/PROSPERO/view/CRD42024599817>); DOI of the previous review: 10.1002/14651858.CD012161.

PLAIN LANGUAGE SUMMARY

Are faster-acting types of insulin more useful than regular human insulin for adults with type 1 diabetes on multiple daily injections?

Key messages

- Compared to regular human insulin, five faster-acting insulin analogues may offer similar benefits in controlling blood glucose levels.
- Compared to regular human insulin, some of them (ultra-rapid lispro, fast-acting aspart and aspart) may reduce the risk of severe complications such as episodes of low blood glucose.
- The evidence about mild diabetes complications, quality of life and adherence to treatment was unclear.

What is type 1 diabetes?

Type 1 diabetes is a condition in which a person's blood sugar (glucose) level becomes too high because the pancreas cannot produce any insulin to control the levels of glucose.

How is the condition treated?

People with type 1 diabetes have to inject insulin to control their glucose levels and stay well. Faster-acting insulin analogues are modified versions of human insulin that are designed to act more quickly after injection than regular human insulin. Examples include insulin lispro, insulin aspart, insulin glulisine, fast-acting insulin aspart (FIAsp) and ultra-rapid lispro insulin (URLi). They can be injected immediately before meals and lead to lower blood sugar levels after food intake.

What did we want to find out?

We wanted to compare the effects of long-term treatment with fast (and very fast)-acting insulin analogues to regular human insulin or another fast-acting insulin in people with type 1 diabetes on multiple daily injections. We wanted to find out about the effects on blood sugar control, episodes of low blood sugar, low blood sugar at nighttime, a serious complication (diabetic ketoacidosis), quality of life and adherence to treatment.

What did we find?

We found 15 studies where people were randomly put into one of two or more treatment groups. The studies compared the faster-acting insulin analogues to regular human insulin and involved 6335 participants. The people in the included studies were monitored (called follow-up) for between 24 and 52 weeks.

Main results

According to our analyses, five faster-acting insulin analogues may be similar to regular human insulin in terms of short-term (defined as less than one year) glycaemic control, as measured by HbA1c (which reflects average blood sugar levels over the past two to three months). We are uncertain about the effects on control of fasting blood glucose levels, mild or moderate episodes of low blood sugar (called hypoglycaemia), severe or serious episodes of low blood sugar at nighttime (called nocturnal hypoglycaemia), or adherence to the insulin treatment. However, they may be better, especially regarding severe hypoglycaemia or a very dangerous complication with high blood sugar and urine ketones (acids) called diabetic ketoacidosis.

We found no clear evidence for mild or moderate hypoglycaemia at nighttime or health-related quality of life (which is physical, mental, emotional and social health). We also did not find any data for long-term follow-up (defined as longer than one year).

What are the limitations of the evidence?

We have limited confidence in the evidence, mainly because most of the studies were carried out in an unblinded way (where healthcare professionals and participants know which treatment they received). There was also a moderate number of dropouts (more than 5% but less than 20% of the participants did not continue the treatment until the end of the study) and there was not enough study detail provided, so that we can't be confident in all the studies. Furthermore, several studies showed inconsistencies in the reporting of their methods and results.

How up to date is this evidence?

This evidence is current as of May 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings (network meta-analysis and CINeMA for glycaemic control)

Patient or population: adults with type 1 diabetes mellitus

Setting: outpatient, multiple daily injections

Intervention: (ultra-)short-acting insulin analogues

Comparison: regular human insulin

Outcome: **glycaemic control: haemoglobin A1c (HbA1c)**

Measured by: lab blood test

Follow-up: 26 weeks to 52 weeks

14 studies 6039 participants	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Certainty of the evidence (CINeMA)
	Risk with regular human insulin	Risk with insulin analogues		
Aspart (mixed estimate)	The median HbA1c was 7.97%	MD -0.14% (-0.21 to -0.06)	-	⊕⊕○○ Low ^{a,b}
Lispro (mixed estimate)		MD -0.22% (-0.46 to 0.02)	-	⊕⊕○○ Low ^{b,c}
Glulisine (indirect estimate)		MD -0.21% (-0.48 to 0.06)	-	⊕⊕○○ Low ^{b,c}
FIAsp (indirect estimate)		MD -0.17% (-0.28 to -0.05)	-	⊕⊕○○ Low ^{a,b}
URLi (indirect estimate)		MD -0.25% (-0.51 to 0.00)	-	⊕⊕○○ Low ^{b,d}

Outcome: **glycaemic control: fasting blood glucose (FBG)**

Measured by: measured by glucometer

Follow-up: 24 weeks to 52 weeks

4 studies 2737 participants	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Certainty of the evidence (CINeMA)
	Risk with regular human insulin	Risk with insulin analogues		
Aspart (mixed estimate)	The median FBG was 8.01 mmol/L	MD 0.80 mmol/L (-0.01 to 1.61)	-	○○○○ Extremely low ^{a,e}
FIAsp (indirect estimate)		MD 0.69 mmol/L (-0.31 to 1.69)	-	○○○○ Extremely low ^{a,e}

Lispro, glulisine, URLi

These interventions could not be assessed as they were not part of the network assessing RHI.

^aDowngraded one level due to within-study bias: the majority of the evidence had major concerns regarding risk of bias.^bDowngraded one level due to incoherence: treatment effects were only from direct or indirect estimates.^cDowngraded one level due to within-study bias and heterogeneity: the majority of the evidence had some concerns.^dDowngraded one level due to within-study bias and imprecision: the majority of the evidence had some concerns regarding risk of bias and the prediction interval from one comparison crosses the minimally important difference.^eDowngraded three levels due to imprecision, heterogeneity and incoherence: unavailable for assessment because few studies were included.**Summary of findings 2. Summary of findings (network meta-analysis and CINeMA for hypoglycaemia)****Patient or population:** adults with type 1 diabetes mellitus**Setting:** outpatient, multiple daily injections**Intervention:** (ultra-)short-acting insulin analogues**Comparison:** regular human insulinOutcome: **mild/moderate (non-severe) hypoglycaemia**

Measured by: clinical standard

Follow-up: 6 months to 52 weeks

6 studies 3967 participants	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Certainty of the evidence (CINeMA)
	Risk with reg- ular human in- sulin	Risk with insulin ana- logues		
Aspart (mixed estimate)	717 per 1000	753 per 1000 (695 to 810)	RR 1.05 (0.97 to 1.13)	⊕○○○ Very low ^{a,b,f}
FIAsp (indirect estimate)		746 per 1000 (688 to 803)	RR 1.04 (0.96 to 1.12)	⊕○○○ Very low ^{a,b,f}
Lispro, glulisine and URLi	These interventions could not be assessed as they were not part of the network assessing RHI.			

Outcome: **severe/serious hypoglycaemia**

Measured by: clinical standard

Follow-up: 26 weeks to 52 weeks

11 studies 5390 participants	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Certainty of the evidence (CINeMA)
	Risk with reg- ular human in- sulin	Risk with insulin ana- logues		
Aspart (mixed estimate)	152 per 1000	135 per 1000 (110 to 167)	RR 0.89 (0.73 to 1.09)	⊕○○○ Very low ^{a,b,g}
Lispro		154 per 1000	RR 1.01	⊕⊕○○

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(mixed estimate)	(78 to 301)	(0.51 to 1.98)	Low ^{b,g,h}
Glulisine (indirect estimate)	234 per 1000 (70 to 789)	RR 1.54 (0.46 to 5.19)	⊕⊕○○ Low ^{b,g}
FIAsp (indirect estimate)	120 per 1000 (84 to 175)	RR 0.79 (0.55 to 1.15)	⊕○○○ Very low ^{a,b,g}
URLi (indirect estimate)	122 per 1000 (55 to 269)	RR 0.80 (0.36 to 1.77)	⊕⊕○○ Low ^{b,g,h}
Outcome: mild/moderate nocturnal hypoglycaemia Measured by: clinical standard Follow-up: 26 weeks to 52 weeks	The interventions could not be assessed due to no networks.		
Outcome: severe/serious nocturnal hypoglycaemia Measured by: clinical standard Follow-up: 24 weeks to 52 weeks			
4 studies 3229 participants	Anticipated absolute effects* (95% CI)		Relative effect (95% CI) Certainty of the evidence (CINeMA)
	Risk with regular human insulin	Risk with insulin analogues	
Aspart (mixed estimate)	99 per 1000	63 per 1000 (45 to 87)	RR 0.63 (0.46 to 0.88) ⊕○○○ Very low ^{a,b,i}
FIAsp (indirect estimate)		52 per 1000 (26 to 103)	RR 0.52 (0.26 to 1.04) ⊕○○○ Very low ^{a,b,f}
Lispro, glulisine and URLi	These interventions could not be assessed as they were not part of the network assessing RHI.		

^aDowngraded one level due to within-study bias: the majority of the evidence had major concerns regarding risk of bias.

^bDowngraded one level due to incoherence: treatment effects were only from direct or indirect estimates.

^cDowngraded one level due to within-study bias and heterogeneity: the majority of the evidence had some concerns in both domains.

^dDowngraded one level due to within-study bias and imprecision: the majority of the evidence had some concerns regarding risk of bias and the prediction interval from one comparison crosses the minimally important difference.

^eDowngraded three levels due to imprecision, heterogeneity and incoherence: unavailable for assessment because few studies were included.

^fDowngraded one level due to some concerns regarding imprecision and heterogeneity: the confidence interval crosses the minimally important difference; inconsistency exists between the prediction interval and confidence interval in relation to the null effect and minimally important difference.

^gDowngraded one level due to imprecision: the confidence interval crosses the null effect and the minimally important difference for benefit and harm.

^hNot downgraded due to within-study bias, although the majority of the studies had some concerns.

ⁱDowngraded one level due to heterogeneity: the prediction interval from one comparison crosses the minimally important difference.

Summary of findings 3. Summary of findings (network meta-analysis and CINeMA for diabetic ketoacidosis (DKA), quality of life and adherence)

Patient or population: adults with type 1 diabetes mellitus

Setting: outpatient, multiple daily injections

Intervention: (ultra-)short-acting insulin analogues

Comparison: regular human insulin

Outcome: **DKA (diabetic ketoacidosis)**

Measured by: clinical diagnosis

Follow-up: 6 months to 52 weeks

9 studies 4599 participants	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Certainty of the evidence (CINeMA)
	Risk with regular human insulin	Risk with insulin analogues		
Aspart (mixed estimate)	5 per 1000	4 per 1000 (1 to 16)	OR 0.80 (0.20 to 3.11)	⊕○○○ Very low ^{a,b,g}
Lispro (mixed estimate)		4 per 1000 (1 to 27)	OR 0.74 (0.10 to 5.39)	⊕⊕○○ Low ^{b,g,h}
FIAsp (indirect estimate)		6 per 1000 (1 to 56)	OR 1.18 (0.12 to 11.24)	⊕⊕○○ Low ^{b,g,h}
URLi (indirect estimate)		4 per 1000 (0 to 83)	OR 0.74 (0.03 to 16.64)	⊕⊕○○ Low ^{b,g,h}
Glulisine	These interventions could not be assessed as they were not part of the network assessing RHI.			
Outcome: quality of life Measured by: clinical standard Follow-up: 6 months to 52 weeks	The interventions could not be assessed due to no networks.			

Outcome: **adherence to treatment**

Measured by: number of participants compliant with trial procedures

Follow-up: 6 months to 52 weeks

5 studies 3579 participants	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Certainty of the evidence (CINeMA)
	Risk with regular human insulin	Risk with insulin analogues		
Aspart	920 per 1000	927 per 1000 (874 to 982)	RR 1.01 (0.95 to 1.07)	⊕○○○ Very low ^{a,b,i}

(mixed estimate)			
FIAsp			
(indirect estimate)	924 per 1000 (865 to 990)	RR 1.01 (0.94 to 1.08)	⊕○○○ Very low ^{a,b,i}
Lispro, glulisine and URLi	These interventions could not be assessed as they were not part of the network assessing RHI.		

^aDowngraded one level due to within-study bias: the majority of the evidence had major concerns regarding risk of bias.

^bDowngraded one level due to incoherence: treatment effects were only from direct or indirect estimates.

^cDowngraded one level due to within-study bias and heterogeneity: the majority of the evidence had some concerns in both domains.

^dDowngraded one level due to within-study bias and imprecision: the majority of the evidence had some concerns regarding risk of bias and the prediction interval from one comparison crosses the minimally important difference.

^eDowngraded three levels due to imprecision, heterogeneity and incoherence: unavailable for assessment because few studies were included.

^fDowngraded one level due to some concerns regarding imprecision and heterogeneity: the confidence interval crosses the minimally important difference; inconsistency exists between the prediction interval and confidence interval in relation to the null effect and minimally important difference.

^gDowngraded one level due to imprecision: the confidence interval crosses the null effect and the minimally important difference for benefit and harm.

^hNot downgraded due to within-study bias, although the majority of the studies had some concerns.

ⁱDowngraded one level due to heterogeneity: the prediction interval from one comparison crosses the minimally important difference.

BACKGROUND

Description of the condition

Type 1 diabetes mellitus is a metabolic disorder caused by a cell-mediated autoimmune destruction of pancreatic β cells. The resulting deficiency in insulin secretion in turn leads to chronic hyperglycaemia (i.e. elevated levels of plasma glucose). To date, there is no cure, and treatment consists of life-long insulin replacement to control blood sugar levels. Long-term complications of diabetes mellitus include retinopathy, nephropathy, neuropathy and increased risk of cardiovascular disease [1, 2, 3].

Description of the intervention and how it might work

Blood sugar control through insulin therapy is the main priority of therapy for people with type 1 diabetes. Since the Diabetes Control and Complications Trial (4), intensive insulin therapy using a basal-bolus regimen has often been the standard. In this regimen, a bolus of insulin is injected before each meal, whereas a longer-acting insulin type is injected once or twice daily. The bolus insulin can either be regular human insulin (RHI) or a(n) (ultra-)short-acting insulin analogue, and the basal insulin can either be neutral protamine Hagedorn (NPH) insulin or a(n) (ultra-)long-acting insulin analogue. Alternatively, people can use insulin pumps, in which (ultra-)short-acting insulin can be injected as a bolus or continuously in very small amounts, so that no special long-acting insulin component is necessary. Nonetheless, the overall uptake of insulin pumps as a primary form of insulin delivery worldwide has been low [5].

In contrast to human endogenous insulin, insulin analogues have a modified molecular structure resulting in different pharmacokinetic profiles. When RHI is injected subcutaneously, the plasma insulin concentration peaks about two to four hours after injection, unlike the much earlier plasma insulin peak in people without diabetes after meal ingestion. This low rise to peak insulin concentration makes it difficult to mimic physiological temporal insulin profiles and is likely to account for much of the observed hyperglycaemia following meals in people with diabetes [6]. Furthermore, since the action of RHI lasts for about six to eight hours, with a peak at about two to four hours, people run the risk of experiencing late post-absorptive hypoglycaemic episodes [7, 8] (Vignati 1997 [9]). The delay in the absorption of subcutaneously administered regular insulin is due to the fact that in this preparation, insulin tends to associate in 'clusters' of six molecules (hexamers), and time is needed after injection for these clusters to dissociate to single molecules that can be used by the body [10]. Short-acting insulin analogues with less tendency toward self-association are therefore absorbed more quickly, achieving peak plasma concentrations about twice as high and within approximately half the time compared to regular insulin [11, 12]. In recent years, ultra-short-acting or ultra-rapid-acting insulins have attracted heated attention due to their swift regulation of glucose levels approximately 10 to 20 minutes after injection and lasting for up to five hours [13]. Together with real-time glucose monitoring techniques, this faster class of insulins initiates real-time measurement for optimal diabetes management [14, 15].

Currently, there are three different short-acting insulin analogues available, including insulin aspart, insulin glulisine and insulin

lispro, and two types of ultra-short-acting insulin analogues, including fast-acting insulin aspart (faster aspart) or ultra-rapid lispro [16]. Compared to RHI, insulin aspart has aspartic acid instead of proline at position 28 of the B-region; in glulisine, the amino acid asparagine is replaced by lysine at position three and lysine with glutamic acid at position 29 of the B-chain; and in insulin lispro, proline at position 28 and lysine at position 29 of the B-region are interchanged. Faster aspart has two additional excipients, including niacinamide (vitamin B₃) and L-arginine [17], whereas ultra-rapid lispro harbours two other excipients, including citrate and treprostinil [18].

Adverse effects of the intervention

The key risk associated with any insulin therapy is the occurrence of hypoglycaemic episodes. Insulin analogues have been promoted as lowering the risk of hypoglycaemia because their faster pharmacokinetic profile might help avoid hypoglycaemic episodes in post-meal periods. However, the evidence needs to be carefully evaluated, also considering different patient subgroups and the methodological challenges associated with assessing hypoglycaemia in clinical trials. For example, Singh et al. point out that several trials on insulin analogues have excluded participants with a history of severe hypoglycaemia [19]. Open-label designs, combined with measurements of hypoglycaemia that rely solely on participants' reports, may yield results at high risk for bias.

Another potential adverse effect of insulin analogues is weight gain. In general, improvement in glycaemic control through insulin therapy is frequently associated with weight gain, which in turn can have negative consequences for blood pressure and lipid profiles [20].

Finally, the structural homology of insulin analogues to insulin-like growth factor I (IGF-I) has caused concern regarding the progression of late diabetic complications and potential mitogenic (induction of cell division) effects, especially with long-term use of insulin analogues. IGF-I may affect the progression of retinopathy [21, 22], and certain modified insulin analogues have shown a carcinogenic effect in the mammary glands in female rats [23], or mitogenic potency in osteosarcoma cells [24]. This has also been explored using observational data [25].

Overall, limited data on the long-term safety of insulin analogues are currently available, primarily due to short follow-up periods and the exclusion of individuals with clinically relevant complications from clinical studies.

Due to their faster pharmacokinetics, insulin analogues could lead to lower glucose levels after meals [11, 26], and potentially also improve overall glycaemic control. It has been proposed that lower post-prandial glucose may be associated with a lower risk of cardiovascular complications in diabetes [27].

Furthermore, insulin analogues might have additional beneficial effects on people's health-related quality of life by requiring less restrictive mealtime planning. For people treated with RHI, insulin should be administered at least 30 minutes before meals. However, this recommendation is often not followed because of its inconvenience [28]. (Ultra-)short-acting insulin analogues, in contrast, can be injected directly before meals or even after meals without a deterioration of prandial glycaemic control [29, 30]. Ultra-rapid analogues of insulin aspart and lispro aim to provide

a faster onset of action and reduce the peak of post-prandial hyperglycaemia [31].

Why it is important to do this review

The pharmaceutical industry has heavily promoted insulin analogues. Based on their pharmacokinetic profile, we might expect (ultra-)short-acting insulin analogues to improve the insulin therapy of people with diabetes mellitus. The evidence collected in previous reviews and meta-analyses has shown at best only modest benefits for glycaemic control and the frequency of hypoglycaemic episodes compared to therapy with RHI [19, 32, 33, 34]. While some reviews find a more substantial reduction in glycosylated haemoglobin A1c (HbA1c) with rapid-acting insulin analogues compared to RHI, the effects were smaller than published minimal clinically relevant differences. Furthermore, potential adverse effects of treatment with these insulin analogues have not been ruled out sufficiently and there is a lack of evidence regarding the effects on long-term clinical outcomes [19, 34]. Treatment effects have not been compared among all different classes of (ultra-)short-acting insulin analogues.

Based on the results of cost-effectiveness analyses [35, 36], the heavy use of insulin analogues promoted through the aggressive marketing of the pharmaceutical industry has become a matter of political debate [37, 38, 39, 40]. Although the production prices are similar, the costs of rapid-acting insulins are higher than regular human insulins [41]. This issue is of particular importance for low- to middle-income countries and low-resource settings, where people still die due to the lack of affordable insulin [38, 42, 43]. Moreover, the usage of insulin pumps can exacerbate the financial burden compared to multiple daily injections [44], and the overall uptake worldwide has been low [5].

The last Cochrane review, published in 2016, indicated a small difference favouring short-acting analogues compared to regular insulin for glycaemic control, with inconclusive data on hypoglycaemia [45]. Since the publication of this review, new trials of ultra-short-acting insulin types have emerged, providing additional evidence on this topic. The World Health Organization (WHO) commissioned our team to update this review and incorporate measurements of relative effectiveness and simultaneous comparison of different types of (ultra-)short-acting insulins using network meta-analysis.

OBJECTIVES

- Primary objective: To compare the effects of long-term treatment with (ultra-)short-acting insulin analogues to regular human insulin or another (ultra-)short-acting insulin analogue in people with type 1 diabetes on multiple daily injections through a network meta-analysis
- Secondary objective: To obtain an estimate of the relative ranking of these (ultra-)short-acting insulin analogues compared to regular human insulin.

METHODS

We followed the *Cochrane Handbook for Systematic Reviews of Interventions* [46] and the Methodological Expectations of Cochrane Intervention Reviews (MECIR) standards for the conduct of the review [47] and the PRISMA 2020 guidelines for the full report of this review [48].

We adapted the inclusion and exclusion criteria from the previous version of the review [45] without focus on any types of insulin pumps, including automatic insulin delivery, artificial pancreas and continuous subcutaneous insulin infusion [49]. Concomitant interventions had to be the same in both the intervention and comparator groups to establish fair comparisons. The previous version of the review considered the following outcomes: all-cause mortality, macrovascular and microvascular complications, severe hypoglycaemic episodes, glycaemic control, adverse events, health-related quality of life and costs. This version included almost all the same outcomes but rearranged in order based on the priority-setting exercise led by the WHO Guideline Development Group (GDG). We excluded 'costs' as this is beyond the scope of this update.

Compared to our previous protocol [50], we made the following changes:

- We expanded the search strategy for (ultra-)rapid-acting insulin analogues to include: head-to-head comparison trials for glulisine; trials about ultra-rapid lispro; trials about fast-acting insulin aspart; relevant clinical study reports.
- Focusing on the regimen of multiple daily injections, we excluded interventions delivered through insulin pumps, and oral and inhaled insulins. The reason for this was that the WHO guideline is focused on the most commonly used regimen worldwide, which remains multiple daily injections.
- Based on the prioritisation of the WHO Guideline Development Group (GDG) in the update, we formed a broad outcome set of critical and important outcomes: we redefined some of the critical outcomes pertaining to hypoglycaemia and added adherence to treatment.
- Instead of using a traditional risk of bias tool as previously, we applied the revised Cochrane risk of bias 2 tool (RoB 2) for each outcome.
- We performed a novel network meta-analysis using a random-effects model with restricted maximum likelihood (REML) adjustments, surface under the cumulative ranking curve (SUCRA) and P scores for treatment ranking via R [51].
- To judge the certainty of evidence, we used the Confidence in Network Meta-analysis (CINeMA) framework and presented the critical outcomes in three summary of findings tables.

Criteria for considering studies for this review

Types of studies

We included parallel randomised controlled trials (RCTs) and the first phase of trials with a cross-over design, which had a treatment duration of 24 weeks or more, designed to compare people with type 1 diabetes who were treated with the currently 'on the market' available short-acting insulin analogues, such as lispro, aspart and glulisine, and ultra-rapid-acting analogues, such as fast-acting insulin aspart and ultra-rapid lispro, versus regular human insulin (RHI) in the review.

We excluded quasi-RCTs, cluster-randomised trials and the second phase of cross-over trials.

Types of participants

Adults (aged 18 years and older) with type 1 diabetes mellitus who were not pregnant.

Types of interventions

We considered all participants with type 1 diabetes receiving a(n) (ultra-)short-acting insulin analogue treatment (intervention group) in comparison to people receiving treatment with RHI (control group), as long as any additional treatment was given equally to both groups.

The short-acting insulins included:

- Glulisine
- Lispro
- Aspart

The ultra-short-acting insulin/ultra-rapid-acting insulin analogues (URAA) included:

- Ultra-rapid lispro insulin (URLi)
- Fast-acting insulin aspart (FIAsp)

We excluded interventions delivered through insulin pumps, and oral and inhaled insulins.

Comparisons and network plotting

We included any comparisons between RHI and any (ultra-)short-acting insulin analogues and also head-to-head comparisons among analogues. For network meta-analysis, we considered the networks including RHI.

Minimum duration of intervention and follow-ups

We considered interventions prescribed for at least 24 weeks after randomisation.

Setting

We considered all settings from primary to specialised care.

Outcome measures

We did not use outcomes as eligibility criteria.

Critical outcomes

- **Glycaemic control:**
 - **HbA1c (haemoglobin A1c):** a measurement of the average blood glucose levels over the past two to three months, expressed as a percentage.
 - **Fasting blood glucose (FBG):** the glucose concentration in the blood after an overnight fast, typically measured in mg/dL or mmol/L.
- **Hypoglycaemia:**
 - **Mild/moderate (non-severe) hypoglycaemia:** episodes where blood glucose levels fall below 70 mg/dL (3.9 mmol/L) but can be self-treated without needing external assistance.
 - **Severe/serious hypoglycaemia:** hypoglycaemia that requires assistance from another person or medical intervention due to confusion, unconsciousness or seizures.
- **Nocturnal hypoglycaemia:** occurring during sleep, often defined as a low blood sugar episode between bedtime and waking, typically with blood glucose levels < 70 mg/dL (3.9 mmol/L).

- **Mild/moderate nocturnal hypoglycaemia:** episodes during sleep that are not life-threatening and can be managed without assistance.
- **Severe/serious nocturnal hypoglycaemia:** nocturnal hypoglycaemia episodes requiring medical assistance or leading to unconsciousness.
- **Diabetic ketoacidosis (DKA):** defined as the acute complication of diabetes leading to elevated ketones in the blood, metabolic acidosis and hyperglycaemia. Common diagnostic criteria include the presence of blood ketones and acidosis.
- **Quality of life:** the subjective measure of the participant's well-being, including physical, mental and social aspects, often assessed using validated instruments such as the Diabetes Quality of Life (DQoL) or SF-36 questionnaires.
- **Adherence to treatment:** the extent to which a participant follows the prescribed insulin regimen. This can be measured through self-reports, pharmacy refill data or electronic monitoring devices.

Important outcomes

- **Death (all-cause mortality):** the number of deaths due to any cause during the study period.
- **Macrovascular complications:**
 - **Myocardial infarction (MI):** usually confirmed by elevated cardiac enzymes, imaging studies and/or electrocardiogram (ECG) changes.
 - **Stroke:** any ischaemic or haemorrhagic stroke confirmed through neuroimaging (computed tomography (CT) or magnetic resonance imaging (MRI)) or clinical diagnosis, with persistent neurological symptoms.
- **Microvascular complications:**
 - **Nephropathy:** kidney damage due to diabetes, commonly measured by albuminuria (urinary albumin excretion rate) and reduced glomerular filtration rate (GFR). Defined as albuminuria > 30 mg/g creatinine or GFR < 60 mL/min/1.73 m².
 - **Retinopathy:** damage to the retina caused by diabetes, assessed via retinal examination or imaging, often graded by severity (e.g. non-proliferative, proliferative).
- **Accidents:** injuries or incidents when using the insulin preparations, typically measured by self-report or medical records.
- **Adverse events:** any unintended or harmful outcome related to the treatment, including both minor and serious events, reported during the trial period.
- **Body weight change:** the difference in body weight from baseline, often measured in kilograms (kg) or as a percentage change, which may indicate the effect of treatment on metabolism and overall health.

Timing of outcome measurement

For all outcome measures, we defined short-term follow-up as up to 12 months and long-term follow-up as more than 12 months.

Minimal clinically important differences (MCID)

We defined the MCID for each critical and important outcome to aid in rating the certainty of the evidence regarding imprecision (see [Supplementary material 7](#)).

Search methods for identification of studies

Electronic searches

We searched the following sources from inception to 14 May 2025, with no restrictions on the language of publication:

- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 4) via the Cochrane Register of Studies Online (CRSO);
- MEDLINE (Ovid MEDLINE ALL 1946 to Daily Update);
- Science Citation Index Expanded and Emerging Sources Citation Index (via Web of Science);
- ClinicalTrials.gov (www.clinicaltrials.gov);
- World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (www.who.int/trialsearch).

We did not include Embase in our search, as RCTs indexed in Embase are now prospectively added to CENTRAL via a highly sensitive screening process [52].

For detailed search strategies, see [Supplementary material 1](#).

Searching other resources

In addition to the electronic search, we extensively reviewed reference lists in included studies and any relevant systematic reviews according to the MECIR standards [53].

To identify potentially problematic studies, we undertook several actions. We cross-referenced trial registers (such as ClinicalTrials.gov, WHO ICTRP) to retrieve all relevant reports and compared them to identify any duplicates. We searched the Retraction Watch Database and included errata, corrections or questionable reports for all included trials. In addition, we reviewed the status of trial registrations to identify any studies withdrawn or terminated due to safety or quality issues. If any study was flagged, we excluded them with a clear justification.

Data collection and analysis

Selection of studies

Two review authors (YG, YQM or TS) independently screened the abstract, title, or both, of all records to determine which studies we should assess further. We obtained the full text of all potentially relevant records. We resolved disagreements through consensus or by recourse to a third review author (JF, BB), if these occurred. In case we were unable to resolve a disagreement, we planned to categorise the study as a 'Study awaiting classification' and to contact the study authors for clarification. We presented a 2020 PRISMA flow diagram to show the process of study selection [48]. We listed all articles excluded after full-text assessment in a 'Characteristics of excluded studies' table and provided the reasons for exclusion.

Data extraction and management

Two review authors independently extracted key participant and intervention characteristics from studies that fulfilled our inclusion criteria. We described interventions according to an adapted version of the 'template for intervention description and replication' (TIDieR) checklist [54, 55]. We collected data on efficacy outcomes and adverse events using standardised data extraction sheets. Any disagreements were resolved through discussion or, if necessary, by consulting a third review author. We provided

information, including trial identifiers, for potentially relevant ongoing trials, in the 'Characteristics of ongoing studies' table. We attempted to locate the protocol and clinical study report (CSR) for each included study. We also emailed all authors of included studies, ongoing trials and studies awaiting classification to enquire whether they were willing to answer questions regarding their studies, documenting these surveys. If required, we subsequently sought any relevant missing information on the study from the primary study author(s).

Risk of bias assessment in included studies

We assessed risk of bias using the Cochrane RoB 2 tool [46]. We focused on the assessment of the effect of assignment to the interventions at baseline without modifying the original tool. We assessed the following critical outcomes: glycaemic control via HbA1c and FBG, general or nighttime hypoglycaemia by severity (non-severe/mild/moderate, severe/serious), diabetic ketoacidosis (DKA), quality of life and adherence to treatment. We also evaluated important outcomes, including death, macrovascular complications (MI and stroke), microvascular complications (nephropathy and retinopathy), accidents, adverse events by severity (minor and serious) and body weight change in kg. We assessed outcomes at short-term and long-term, respectively.

We used the following bias criteria:

- bias arising from the randomisation process;
- bias due to deviations from intended interventions;
- bias due to missing outcome data;
- bias in the measurement of the outcome;
- bias in the selection of the reported result.

We judged the risk of bias criteria as 'low risk', 'some concerns' or 'high risk' [56, 57]. We included all studies, regardless of the risk of bias judgement, in the meta-analyses.

Measures of treatment effect

- **Relative treatment effect:** To enhance the interpretability of results, we expressed dichotomous data as risk ratios (RRs) with 95% confidence intervals (CIs). For outcomes with rare or even zero events, we applied inverse-variance log odds ratios (ORs) with a 0.5 continuity correction for zero cells to retain all studies for network meta-analysis [58]. We also applied the Mantel-Haenszel method with ORs for the outcome [59]. We expressed continuous data as mean differences (MDs) with 95% CIs. We used standardised mean differences (SMDs) when pooling data from different scales. We prioritised post-intervention over change from baseline measurements.
- **Relative treatment ranking:** We estimated the ranking probabilities that all treatments would be at each possible rank for each intervention and outcome.
 - We obtained a treatment hierarchy using P scores for the critical outcomes of the review [60]. P scores allow the description of the mean extent of certainty that the underlying treatment effect is larger than that of any other intervention.
 - We also used the surface under the cumulative ranking curve (SUCRA) to rank the effectiveness and safety of the different types of insulin [61, 62]. SUCRA accounts for both effect size magnitude and uncertainty around the underlying effect size. We displayed results (network plot, SUCRA plots and league

table) using the 'netmeta package' in R and Rstudio (version 4.4.3) [51, 58].

Unit of analysis issues

We took into account the level at which the randomisation occurred, such as cross-over trials and multiple observations for the same outcome. For cross-over trials, we included data from the first-period (Ferguson 2001 [63, 64]) or approximated a paired analysis by imputing missing standard deviations (Provenzano 2001 [65]) [66]. If more than one comparison from the same study was eligible for inclusion in the same meta-analysis, we either combined groups to create a single pair-wise comparison, or we appropriately reduced the sample size so that the same participants did not contribute data to the meta-analysis more than once (splitting the 'shared' group into two or more groups). Although the latter approach offered some solutions for adjusting the precision of the comparison, it did not account for correlation arising from the inclusion of the same set of participants in multiple comparisons [67].

Dealing with missing data

We obtained relevant missing data from authors, if feasible, and evaluated important numerical data such as screened, eligible, randomised participants as well as intention-to-treat (ITT), as-treated and per-protocol populations. We investigated attrition rates, for example dropouts, losses to follow-up and withdrawals, and critically appraised issues of missing data and imputation methods (e.g. last observation carried forward (LOCF)).

Where standard deviations for outcomes were not reported, we imputed these values by assuming the standard deviation of the missing outcome to be the mean of the standard deviations from those studies where this information was reported. We investigated the impact of imputation on meta-analyses by means of sensitivity analysis.

Reporting bias assessment

For the pairwise meta-analyses, if we were able to pool 10 studies or more, we created and examined a contour-enhanced funnel plot to explore possible small-study and publication biases for the main comparisons (insulin analogues versus RHI) and the critical outcomes [68].

For the network meta-analyses, if we included 10 studies or more per comparison and outcome, we evaluated the presence of small-study effects by drawing comparison-adjusted funnel plots that accounted for the fact that different studies compared different sets of interventions considering their novelty/approval date [58].

Synthesis methods

We performed a meta-analysis if we identified two or more studies with no important differences in methods, addressing the same outcome and similar participants, interventions and comparators in their underlying clinical questions.

Methods for direct treatment comparisons (pairwise meta-analysis)

We performed analyses according to recommendations provided in Chapter 9 of the *Cochrane Handbook* [67], and we used Cochrane's statistical software, Review Manager, for analysis [69]. When possible, we performed these standard pairwise meta-

analyses using a random-effects model because we anticipated methodological and clinical heterogeneity across studies. We calculated corresponding 95% CIs for all effect estimates in the analyses and graphically presented the results using forest plots. When trials were clinically too heterogeneous to be combined, we performed only subgroup analyses without calculating an overall estimate. In order to avoid duplication with the supporting trials of this network meta-analysis, we described only the pairwise comparisons for the data that could not be included in the network due to concerns about transitivity or disconnection from the network.

Methods for indirect and network comparisons

We fit a random-effects network meta-analysis model because we anticipated methodological and clinical heterogeneity across studies. We assumed a common within-network heterogeneity estimate across comparisons, and we estimated this using the restricted maximum likelihood (REML) method [58]. This is a reasonable assumption, given that all treatments included in the network will be of the same nature. An advantage of this approach is that treatment comparisons informed by a single study can borrow strength from the rest of the studies in the network [70, 71]. Each network meta-analysis treatment effect estimate was presented along with a 95% CI and a 95% prediction interval (Prl) with reference to regular human insulin. A PrI is an interval within which the treatment effect estimate of a future study is expected to lie, accounting for both the uncertainty of the treatment effect and between-study variance estimates [72]. We included arm-based data using regular human insulin as a common comparator to avoid unit of analysis issues when incorporating data from multiple-arm studies. We conducted a network meta-analysis using the network suite of commands in R [51, 58, 73].

Investigation of heterogeneity and subgroup analysis

Assessment of heterogeneity

We assessed between-study heterogeneity through visual inspection of forest plots to assess the overlap of CIs and used the I^2 statistic, which quantifies between-study variation across studies, to assess the impact of heterogeneity on the meta-analysis. For network meta-analysis, we assumed a common within-network heterogeneity estimate across comparisons, and we used REML to estimate the heterogeneity variance and the Hartung and Knapp, Sidik and Jonkman (HKSJ) method for calculating confidence intervals for the summary effect [74, 75]. We interpreted the I^2 statistic as follows [67]:

- 0% to 40%: may not be important;
- 30% to 60%: may indicate moderate heterogeneity;
- 50% to 90%: may indicate substantial heterogeneity;
- 75% to 100%: indicates considerable heterogeneity.

We also used Cochran's Q test to assess for heterogeneity of estimated effect sizes from individual studies and calculate prediction intervals. However, we cautiously interpreted these results considering both the low power to detect true heterogeneity when the number of studies is small and the excessive power needed to detect negligible heterogeneity when the number of studies is high.

Subgroup analysis

We performed subgroup analyses for participants based on the following characteristics:

- Population: 1) age group (adults and elderly adults), 2) baseline Hb1Ac (for instance, < 6.5%, 6.5% to 8%, > 8%) and 3) presence of comorbidities.
- Intervention: 1) short-acting duration, 2) regimen and 3) basal insulin therapy.
- Setting: 1) level of care, 2) low- and middle-income countries (LMIC) versus high-income countries (HIC) [76].

Depending on data availability, we conducted network meta-regression (based on pre-specified effect modifiers) [77].

Assessment of transitivity and inconsistency (network meta-analysis)

- **Assessment of the transitivity assumption:** We assessed the transitivity assumption before conducting a network meta-analysis. Network meta-analysis rests on the assumption of transitivity, that is, that effect modifiers have a comparable distribution across treatment comparisons in a network [78, 79]. To assess the plausibility of this assumption, we visually inspected the comparability of distributions of age groups (adults and elderly adults), baseline Hb1AC, type of basal insulin therapy, short-acting add-on regimes (intensive treatment/corrections) and comorbidities as potential treatment effect modifiers across comparisons. We assessed the similarity of the inclusion and exclusion criteria of all studies, including participants, treatments and outcomes, to evaluate whether they impacted treatment effects.
- **Assessment of statistical consistency:** Lack of transitivity in a network can threaten the validity of the consistency assumption, that is, the statistical agreement between direct and indirect evidence [80, 81]. Results can be misleading in the presence of inconsistency in the network. We evaluated the presence of inconsistency both locally and globally. We evaluated each network locally using the loop-specific method by generating an inconsistency factor along with a 95% CI for each closed loop [82]. In this way, we identified which piece of evidence would be responsible for inconsistency, and we explored this further. We also applied a global assessment for consistency in each network by applying the design-by-treatment interaction model [73]. It has been shown that inconsistency tests have low power to detect true inconsistency [83, 84]. Hence, we assessed transitivity even in the absence of evidence for inconsistency. If inconsistency was found, we followed the guidance provided in the *Cochrane Handbook* Section 11.4.4.4, including pre-specified subgroup analysis, sensitivity analysis and network meta-regression (based on pre-specified effect modifiers) [77].

Equity-related assessment

The *Cochrane Handbook* defines different approaches for assessing health equity [85], of which we applied the following:

- To explore the distribution of effects across one or more PROGRESS-Plus characteristics: We intended to include subgroup analysis to assess the distribution of effects across different ages, genders, ethnicities, settings, socioeconomic status of families and income classifications of countries.
- To assess interventions that focus on disadvantaged or at-risk populations: We did not restrict the inclusion of interventions targeting these subpopulations.

Sensitivity analysis

We performed sensitivity analyses for both pairwise and network meta-analyses by restricting analysis to studies with a low risk of bias.

Certainty of the evidence assessment

We drafted summary of findings tables to summarise the review results for the critical outcomes, using the CINeMA framework and software [86, 87, 88].

Our reference treatment for the network meta-analysis is regular human insulin. We considered the six domains adapted from the GRADE criteria to CINeMA to evaluate the certainty of the body of evidence as it relates to studies that contributed data to the network meta-analysis for each pre-specified outcome: (a) within-study bias, (b) reporting bias, (c) indirectness, (d) imprecision, (e) heterogeneity and (f) incoherence [86]. Two review authors independently made judgements about the certainty of the evidence (high, moderate, low or very low) and resolved disagreements by discussion or consultation with a third review author. We created a summary of findings table for each outcome, using the approach presented by Yepes-Nuñez et al. [89].

Consumer involvement

The outcomes for this review were decided through input from two guideline development groups. Both groups included a wide variety of experts, including people with lived experience, nurses, academics and clinicians.

RESULTS

Description of studies

For a detailed description of trials, see [Supplementary material 2](#), [Supplementary material 3](#) and [Supplementary material 4](#) tables.

Results of the search

The electronic search yielded 6325 records. After removing duplicates, we screened 4385 studies and retrieved 123 records for full-text screening. We then included six trials for data extraction. Together with the nine included studies from the previous systematic review [45], we therefore included 15 trials in this review. For further details, see the flow chart in [Figure 1](#).

Figure 1. PRISMA flow diagram

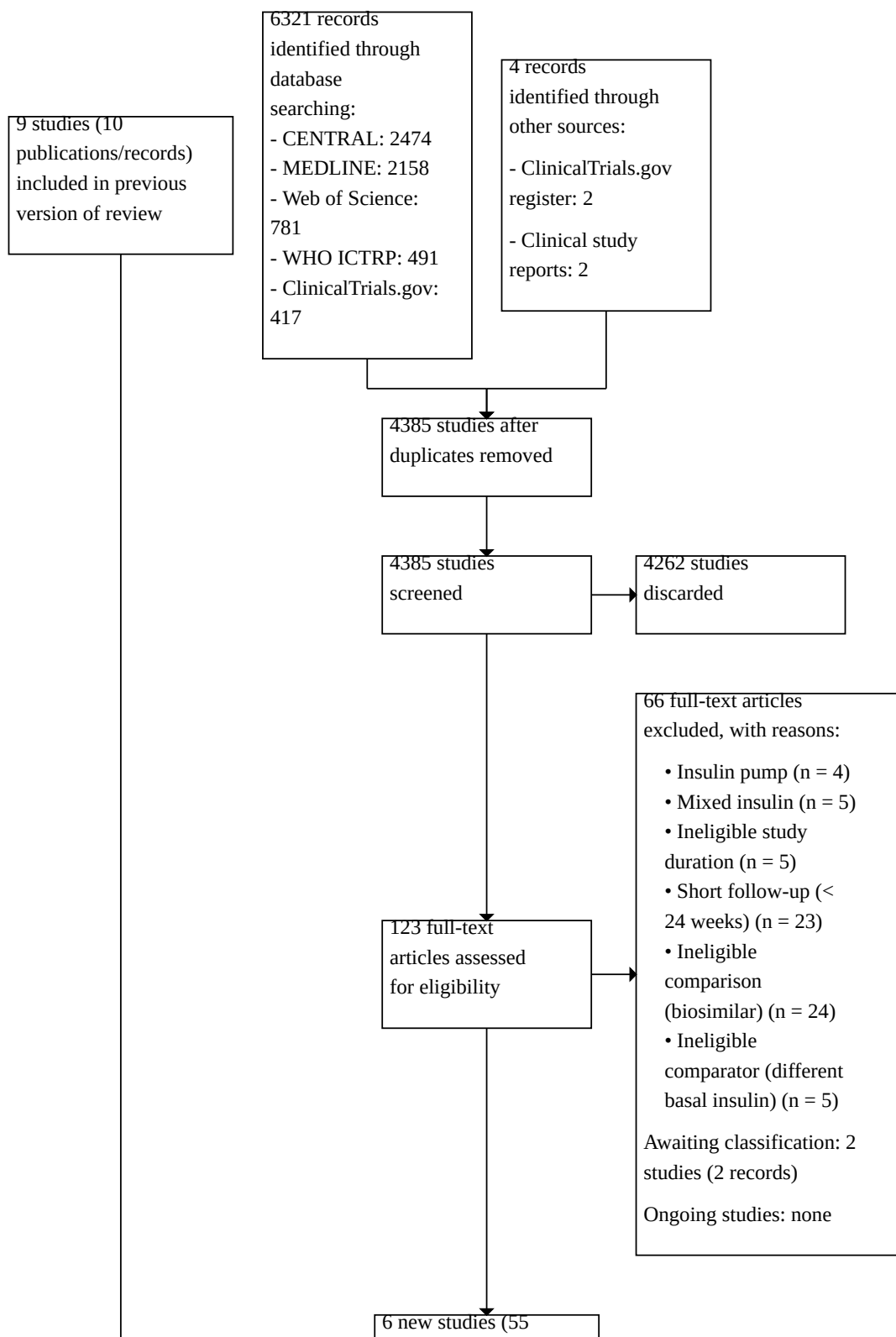
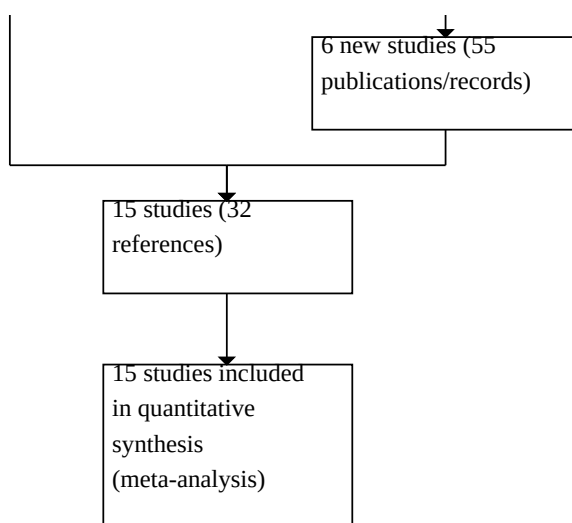


Figure 1. (Continued)



We placed two studies in 'studies awaiting classification' due to lack of further publications or records (NCT00297583 [90]; Schade 1996 [91]).

Included studies

A detailed description of the characteristics of the included trials is in [Table 1](#) and [Table 2](#).

Source of data

The results of all the trials were at least partially published in scientific journals between 1996 and 2022. For six of the trials, we relied on additional information based on the original trial reports, which were published in a report by the Institute for Quality and Efficiency in Health Care in 2007 [92]. Therefore, we cited this report as an additional source for these six trials. The publications by Anderson 1997 [93] and Garg 1996 [94] were based on the combined data of two [93] and three [94] different trials. From the publications alone, it was not clear that the data from different trials were combined. However, the original trial reports were available for the report by the Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen (IQWiG 2007) [92], and we therefore continued to treat these trials separately in this review using the same study names (Z011, Z013 and Z015) as in the report [92]. Previous review authors contacted all trial authors to request missing data or clarify issues regarding the methodology of the included trials. However, only one of the trial authors replied. For this update, we contacted two authors out of the four URAA trials due to missing data, but received replies from neither (Ma 2024 [95, 96, 97]; PRONTO-T1DM [98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118]).

Comparisons

The included trials comprised five pairwise comparisons. Six trials compared the insulin analogue lispro with RHI (Ferguson 2001; Provenzano 2001; Recasens 2003 [119, 120]; Z011 2007 [121, 122, 123]; Z013 2007 [124, 125, 126]; Z015 2007 [127, 128]), three trials compared the insulin analogue aspart with RHI (Home 2000 [129, 130, 131, 132, 133]; Iwamoto 2001 [134]; Raskin 2000 [135, 136]),

two trials used glulisine and lispro (Dreyer 2005 [137]; Kawamori 2009 [138, 139]), two trials used URLi and insulin lispro (Ma 2024; PRONTO-T1DM), and two trials compared FIAsp with insulin aspart (ONSET 1 [140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158]; ONSET 8 [159, 160, 161, 162, 163, 164, 165, 166, 167]).

Overview of study populations

Overall, 6335 people with type 1 diabetes on multiple daily injections participated in the 15 included trials. Except Ferguson 2001, with unclear randomisation in two arms, 5332 participants were randomised to the treatment group receiving an (ultra-)short-acting insulin analogue and 964 participants were randomised to the control group receiving RHI. In the two cross-over trials, 51 participants were in both treatment arms (Ferguson 2001; Provenzano 2001).

The individual sample sizes ranged from 12 to 1070 participants across trials.

Study design and setting

All included trials were RCTs. Thirteen trials used a parallel design and two trials were cross-over studies (Ferguson 2001; Provenzano 2001).

Eleven trials evaluating short-acting insulin analogues were open-label with no blinding of either participants or investigators, but four trials evaluating URAAs used double-blinding.

Among the 11 trials evaluating short-acting insulin analogues, five trials provided no information regarding the years in which the studies were carried out (Dreyer 2005; Ferguson 2001; Iwamoto 2001; Provenzano 2001; Recasens 2003). The other six trials were performed between 1992 and 2006. The four trials evaluating URAAs were launched between 2013 and 2022.

Overall, 10 of the trials were carried out in a multicentre setting, while five trials were single-centre studies. The setting was not reported for two trials (Provenzano 2001; Recasens 2003). Seven

of the multicentre trials reported the number of study centres involved, which were 30 (Ma 2024), 59 (Raskin 2000), 67 (Dreyer 2005), 88 (Home 2000), 146 (ONSET 8), 165 (ONSET 1) and 166 centres (PRONTO-T1DM).

All trials were either funded commercially (Dreyer 2005; Ferguson 2001; Home 2000; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Raskin 2000; Z011 2007; Z013 2007; Z015 2007), or the funding was not reported (Provenzano 2001; Recasens 2003).

Two trials were carried out in Japan (Iwamoto 2001; Kawamori 2009), while the rest of the trials were predominantly carried out in North America and Europe, but five trials also included study centres in South Africa, Australia (Z011 2007; Z013 2007) and Asia (Ma 2024; ONSET 8; PRONTO-T1DM).

Ten multicentre trials had an outpatient setting (Home 2000; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Raskin 2000; Z011 2007; Z013 2007; Z015 2007). The other trials, even though not always explicitly stated, can be assumed to have also been carried out in an outpatient setting.

The duration of intervention ranged from 24 to 52 weeks, with a mean of about 34 weeks. Seven of the trials reported a run-in period lasting from two to eight weeks to achieve stable metabolic conditions. None of the trials had to be terminated before the planned end of follow-up.

Participants

There was some diversity of the participant populations included in the different trials. For example, one trial only included participants with an impaired awareness of hypoglycaemia who had been diagnosed with type 1 diabetes for at least five years (Ferguson 2001), while another trial only included people who were newly diagnosed (Recasens 2003).

Overall, the weighted mean age of the participants was 36.4 years with the mean age ranging between 23 and 46.1 years across trials. Female percentages ranged from 33 to 82 and the mean body mass was 24.3 kg/m² with the trial means ranging from 21 kg/m² to 26.7 kg/m².

The mean disease duration across trials ranged from 8.1 weeks to 26 years with a mean disease duration of all participants of 13.1 years. Whereas the majority of the trials enrolled participants with over 10 years' diabetes, two trials enrolled participants diagnosed for less than half a year (Recasens 2003; Z015 2007). The participants' mean HbA1c was 8.1%, but the trials' mean baseline HbA1c varied between 7.5% and 11.4%. The trials did not report data on disease severity.

Most trials reported co-medications, such as once or twice daily long-acting insulins, including glargine, degludec, detemir, NPH and ultralente. Two trials used intermediate insulins (Ferguson 2001; Iwamoto 2001), whereas one trial did not state the name of the short-acting insulin (Provenzano 2001). Ma 2024 also reported the use of metformin among 4% to 5.1% of participants in each arm. Only ONSET 1 and ONSET 8 reported multiple comorbidities, such as diabetes-related complications, dyslipidaemia, hypertension and depression.

Eight trials provided information on race (Home 2000; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Raskin 2000). In Iwamoto 2001 and Kawamori 2009, all participants were Asian. In Home 2000, 99% of participants were white and in Raskin 2000, 94% of the participants were white. Among the four trials with ultra-rapid insulin analogues (Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM), three enrolled participants who were white, whereas Ma 2024 recruited 80% Asians. Three trials reported ethnicity, among which approximately 90% were 'not Hispanic or Latino' (ONSET 1; ONSET 8; PRONTO-T1DM).

The criteria for entry into the individual trials are outlined in [Table 1](#) and [Table 2](#). Major exclusion criteria were insulin pump therapy and advanced diabetic complications.

Interventions

All trials tried to apply a comparable insulin regimen throughout the investigation period but, usually, insulin therapy was left somewhat flexible with the aim of reaching the best possible glycaemic control. Most trials used NPH, glargine, degludec or detemir as basal insulin, one trial used ultralente insulin (Z011 2007), and another trial allowed both, NPH or ultralente insulin (Z015 2007). Three trials did not specify the type of basal insulin (Ferguson 2001; Iwamoto 2001; Provenzano 2001). Eleven studies of short-acting insulins were injected pre-prandially, whereas the four trials of URAs were used prandially. Among the latter four trials, two designed an open-label arm for post-prandial use, but failed to run the trial until the final endpoint (ONSET 1; PRONTO-T1DM). We also did not include the open arm for post-prandial use from ONSET 8 due to the scarcity of data.

All but two trials reported on the treatment before the start of the trial: in Home 2000, Iwamoto 2001 and Raskin 2000, participants had been treated with insulin for at least one year; in Provenzano 2001, Z011 2007, Z013 2007 and Z015 2007, participants had received insulin treatment for at least two months. Fourteen trials reported a "run-in" period of four to eight weeks (Dreyer 2005; Ferguson 2001; Home 2000; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Provenzano 2001; Raskin 2000; Z011 2007; Z013 2007; Z015 2007).

Outcomes

Outcome data could not be extracted from one trial due to a lack of data before cross-over (Ferguson 2001), whereas the other 14 trials had outcomes that we could extract. Only five trials clearly defined a primary study outcome (Provenzano 2001; Recasens 2003; Z011 2007; Z013 2007; Z015 2007). In Ferguson 2001, the primary endpoint was severe hypoglycaemia. For the remaining trials it was glycaemic control using HbA1c (Dreyer 2005; Home 2000; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Raskin 2000). The trials Z011 2007, Z013 2007 and Z015 2007 provided inconsistent information regarding the primary study endpoints. The original study reports referred to "postprandial blood glucose levels" as the "primary efficacy variable" while the study protocol referred to the variables "postprandial glucose excursions", "hypoglycaemia episodes in relation to glycaemic control" and "metabolic control" as "primary efficacy variables". Furthermore, the power analysis was carried out based on the variables pre-prandial blood glucose, HbA1c and hypoglycaemia. Seven of the trials explicitly defined secondary outcomes (Dreyer 2005; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; ONSET

8; PRONTO-T1DM). The remaining trials did not explicitly specify a secondary study endpoint.

Funding

Thirteen of the included trials were funded commercially, including five by Novo Nordisk A/S (Home 2000; Iwamoto 2001; ONSET 1; ONSET 8; Raskin 2000) and six by Eli Lilly (Ferguson 2001; Ma 2024; PRONTO-T1DM; Z011 2007; Z013 2007; Z015 2007). One was sponsored by Aventis (Dreyer 2005) and another one by Sanofi (Kawamori 2009). Only two trials did not state their funding resources (Provenzano 2001; Recasens 2003). More detailed information is provided in [Supplementary material 10](#).

Excluded studies

Overall, we excluded 66 records after full-text screening. Reasons for exclusion of records are given in the 'Characteristics of excluded studies' table ([Supplementary material 3](#)). The main reasons for exclusion were use of insulin pump, mixed insulin, inappropriate study duration, shorter than 24 weeks' follow-up, inappropriate comparator (different basal insulin) and inappropriate comparison (biosimilar) (for details see [Figure 1](#)).

Studies awaiting classification

Two studies could not be retrieved for further data (NCT00297583; Schade 1996).

Risk of bias in included studies

For this update, we used the Cochrane RoB 2 tool. We summarised our judgements and supporting statements for the five RoB 2 domains and made an overall assessment for each outcome described in each of the included trials.

Among the 14 trials assessing glycaemic control measured by HbA1c, we rated one as having overall low risk of bias (ONSET 8), whereas 11 had some concerns overall (Dreyer 2005; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; PRONTO-T1DM; Provenzano 2001; Recasens 2003; Z011 2007; Z013 2007; Z015 2007); two had high risk of bias (Home 2000; Raskin 2000).

Among the four trials involving fasting blood glucose, we rated three as having an overall low risk of bias (Ma 2024; ONSET 1; ONSET 8), whereas one had some concerns overall (Home 2000).

Among the six trials assessing mild to moderate (non-severe) hypoglycaemia, ONSET 8 was at an overall low risk of bias, whereas three had some concerns overall (Ma 2024; ONSET 1; PRONTO-T1DM); two were at a high risk of bias (Home 2000; Iwamoto 2001).

Regarding severe/serious hypoglycaemia, ONSET 8 had low risk of bias, whereas we rated eight trials as having some concerns overall (Kawamori 2009; Ma 2024; ONSET 1; PRONTO-T1DM; Recasens 2003; Z011 2007; Z013 2007; Z015 2007); two trials had high risk of bias (Home 2000; Raskin 2000).

We rated four trials assessing mild to moderate nocturnal hypoglycaemia as having an overall low risk of bias (Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM).

Regarding severe/serious nocturnal hypoglycaemia, two trials had low risk of bias (ONSET 1; ONSET 8), whereas we rated the other two trials as having some concerns overall (Home 2000; Raskin 2000).

Among the nine trials assessing DKA, we rated six as having some concerns overall (Ma 2024; ONSET 1; PRONTO-T1DM; Z011 2007; Z013 2007; Z015 2007), whereas three had overall high risk of bias (Home 2000; Iwamoto 2001; Raskin 2000).

Regarding the two trials assessing quality of life, ONSET 1 had some concerns overall, whereas Home 2000 had a high risk of bias.

Among the five trials measuring adherence to treatment, one had a low risk of bias (ONSET 8), whereas we rated three as having some concerns overall (Ma 2024; ONSET 1; PRONTO-T1DM); Raskin 2000 had a high risk of bias.

Among the 11 trials assessing all-cause mortality, we rated one as having a low risk of bias (ONSET 8), whereas seven had some concerns overall (Ma 2024; ONSET 1; PRONTO-T1DM; Recasens 2003; Z011 2007; Z013 2007; Z015 2007). We rated three as having a high risk of bias (Home 2000; Iwamoto 2001; Raskin 2000).

Among the four trials involving non-fatal myocardial infarction and stroke, ONSET 8 had a low risk of bias, whereas we rated three as having some concerns (Ma 2024; ONSET 1; PRONTO-T1DM).

Among the three trials involving nephropathy, ONSET 8 had a low risk of bias, whereas Ma 2024 and ONSET 1 had some concerns overall.

For retinopathy, we rated ONSET 8 as having a low risk of bias, whereas ONSET 1 had some concerns overall and Iwamoto 2001 had a high risk of bias.

Regarding the two trials assessing accidents, ONSET 8 had a low risk of bias, whereas we rated ONSET 1 as having some concerns.

Among the eight trials assessing non-serious and serious adverse events, ONSET 8 had a low risk of bias, whereas we rated five as having some concerns overall (Dreyer 2005; Kawamori 2009; Ma 2024; ONSET 1; PRONTO-T1DM); Home 2000 and Iwamoto 2001 had a high risk of bias.

Among the eight trials measuring body weight, we rated ONSET 8 as having a low risk of bias, whereas five had some concerns overall (ONSET 1; Recasens 2003; Z011 2007; Z013 2007; Z015 2007); Home 2000 and Raskin 2000 had a high risk of bias.

Synthesis of results

Baseline characteristics

For details of baseline characteristics, see [Table 1](#) and [Table 2](#). The summary of findings tables for the network meta-analysis are [Summary of findings 1](#), [Summary of findings 2](#) and [Summary of findings 3](#). The summary of findings tables for the pairwise meta-analysis can be found in [Supplementary material 11](#), [Supplementary material 12](#), [Supplementary material 13](#), [Supplementary material 14](#) and [Supplementary material 15](#).

Network meta-analysis of critical outcomes: (ultra-)short-acting insulin analogues versus regular human insulin(RHI)

The geometry of the networks is presented in each of the figures ([Figure 2](#); [Figure 3](#); [Figure 4](#); [Figure 5](#); [Figure 6](#); [Figure 7](#)). Due to the few included trials, the networks were star-shaped without any closed loops and, in some cases, there were disconnected networks, i.e. for mild and severe adverse events. We assessed

the certainty of the evidence according to the six domains of CINeMA and this is presented in the summary of finding tables. The following analyses present data from networks with no concerns

about reporting bias or indirectness. We judged incoherence as having major concerns due to the existence of solely indirect evidence between insulin glulisine, FIAsp and URli.

Figure 2. (A) Visual representation of the network. (B) Comparison-adjusted funnel plot for the detection of publication bias. (C) The surface under the cumulative ranking curve (SUCRA), ranking the effectiveness of different treatments. A higher SUCRA value (from 0 to 100%) indicates a higher probability of the intervention being among the best-performing treatments. (D) Forest plot representing the estimates from the network meta-analysis. Abbreviations: lis: lispro, asp: aspart, rhi: regular human insulin, urlis: ultra rapid lispro, fas_asp: fast-acting aspart

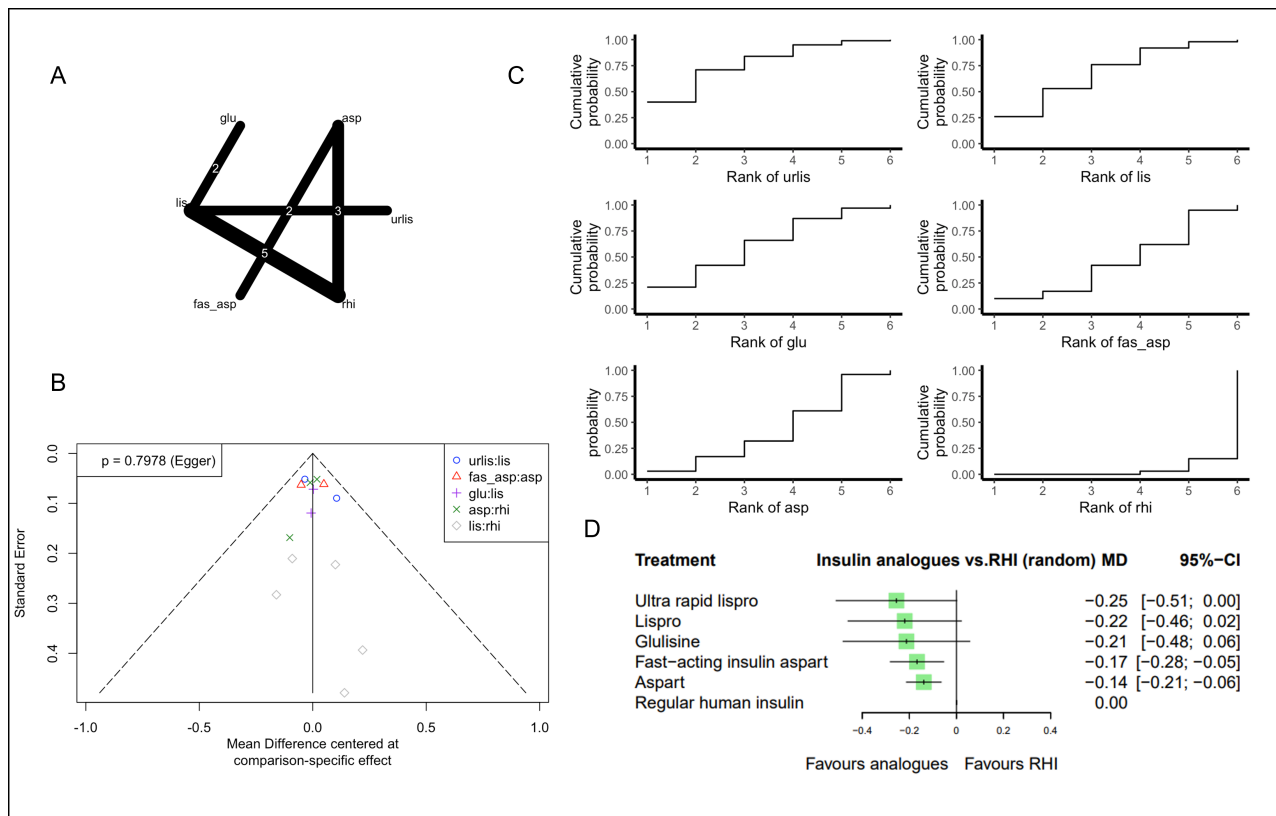


Figure 3. (A) Visual representation of the network among included studies (disconnected network with regular human insulin is not further analysed). (B) Forest plot representing the estimates from the network meta-analysis. (C) The surface under the cumulative ranking curve (SUCRA), ranking the effectiveness of different treatments. A higher SUCRA value (from 0 to 100%) indicates a higher probability of the intervention being among the best-performing treatments. Abbreviations: lis: lispro, asp: aspart, rhi: regular human insulin, urlis: ultra rapid lispro, fas_asp: fast-acting aspart

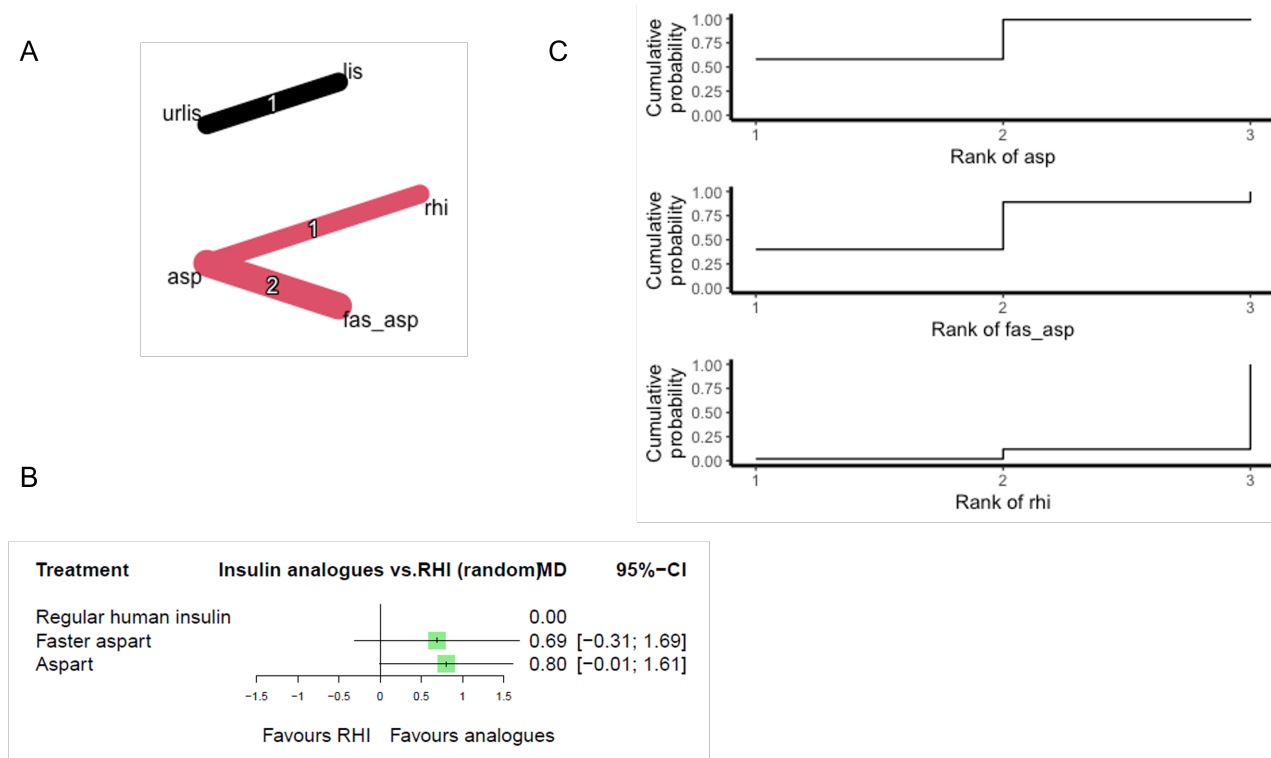


Figure 4. (A) Visual representation of the network among included studies (disconnected network with regular human insulin is not further analysed). (B) Forest plot representing the estimates from the network meta-analysis. (C) The surface under the cumulative ranking curve (SUCRA), ranking the effectiveness of different treatments. A higher SUCRA value (from 0 to 100%) indicates a higher probability of the intervention being among the best-performing treatments. Abbreviations: lis: lispro, asp: aspart, rhi: regular human insulin, urlis: ultra rapid lispro, fas_asp: fast-acting aspart

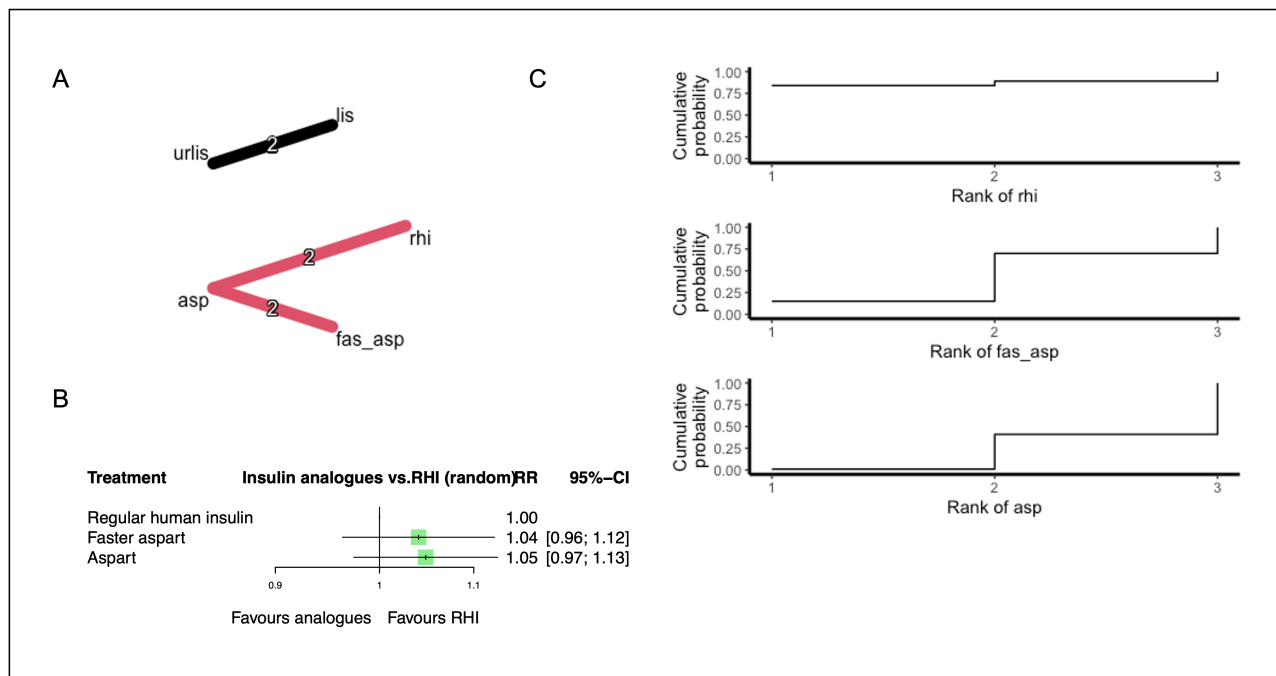


Figure 5. (A) Visual representation of the network. (B) Comparison-adjusted funnel plot for the detection of publication bias. (C) The surface under the cumulative ranking curve (SUCRA), ranking the effectiveness of different treatments. A higher SUCRA value (from 0 to 100%) indicates a higher probability of the intervention being among the best-performing treatments. (D) Forest plot representing the estimates from the network meta-analysis. Abbreviations: lis: lispro, asp: aspart, rhi: regular human insulin, urlis: ultra rapid lispro, fas_asp: fast-acting aspart

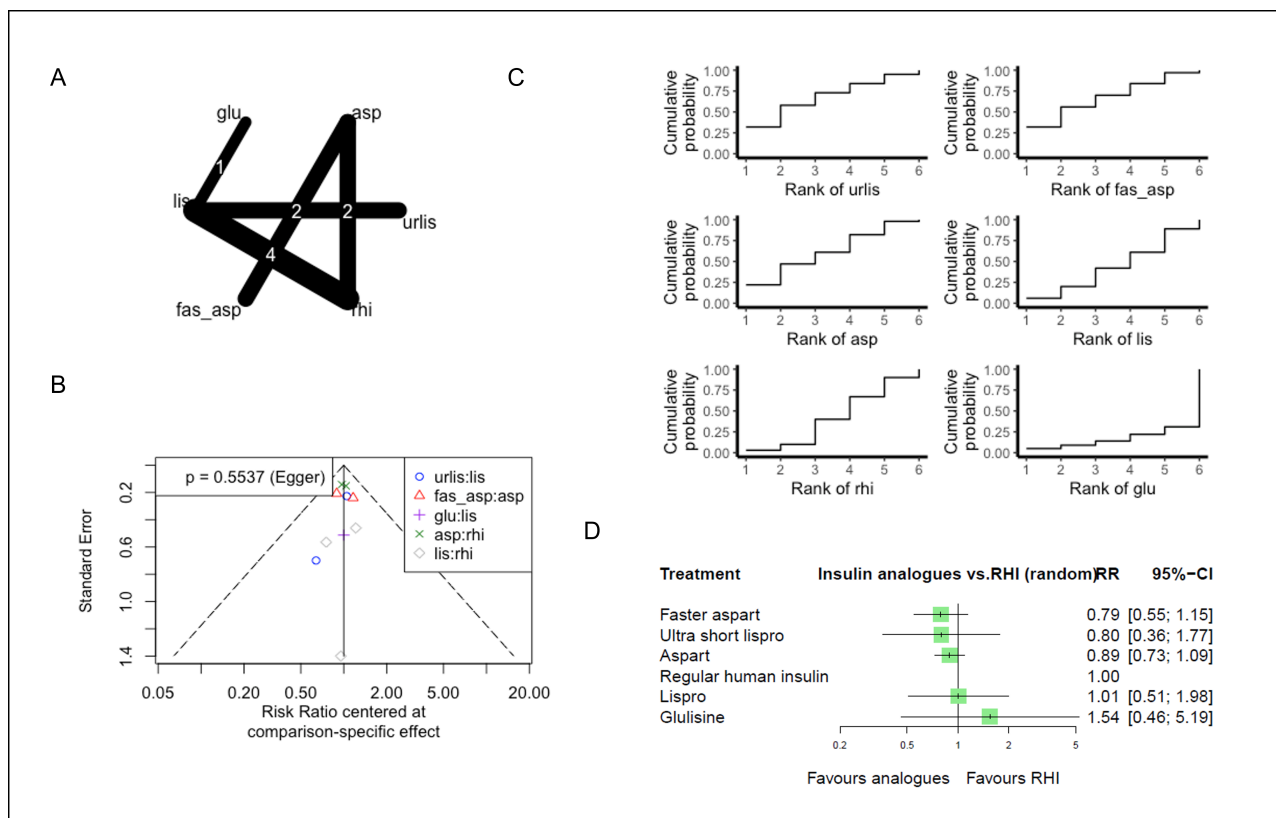


Figure 6. (A) Visual representation of the network. (B) Forest plot representing the estimates from the network meta-analysis. (C) The surface under the cumulative ranking curve (SUCRA), ranking the effectiveness of different treatments. A higher SUCRA value (from 0 to 100%) indicates a higher probability of the intervention being among the best-performing treatments. Abbreviations: lis: lispro, asp: aspart, rhi: regular human insulin, urlis: ultra rapid lispro, fas_asp: fast-acting aspart

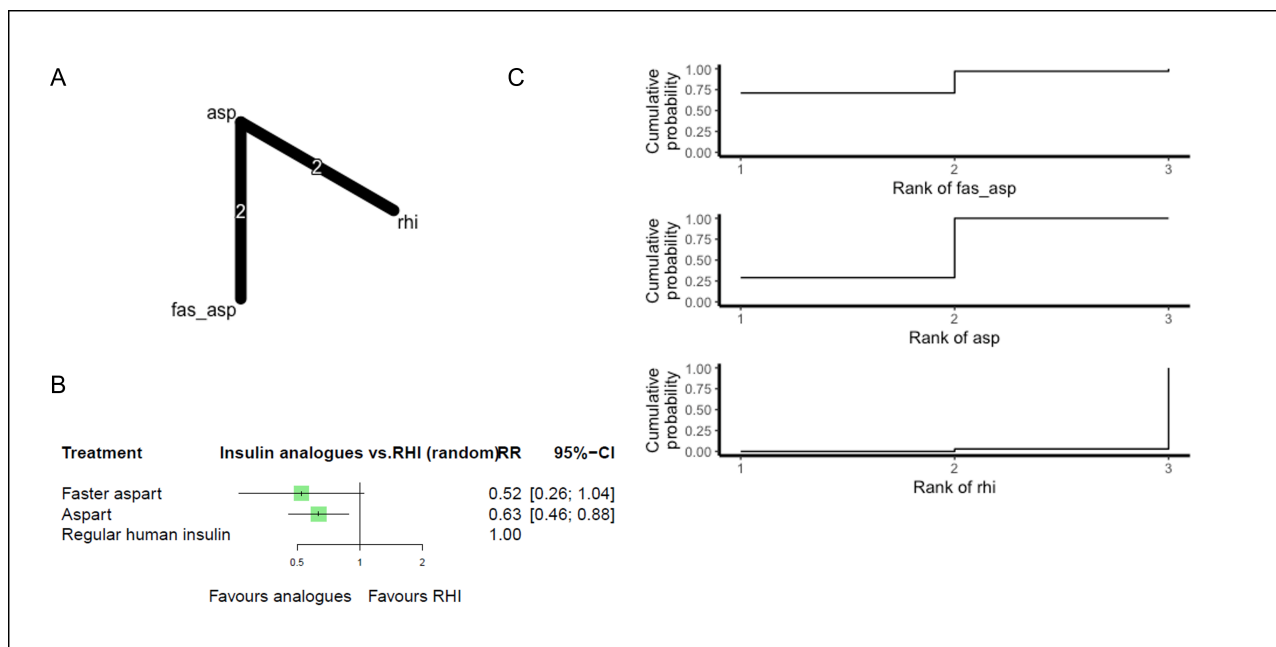
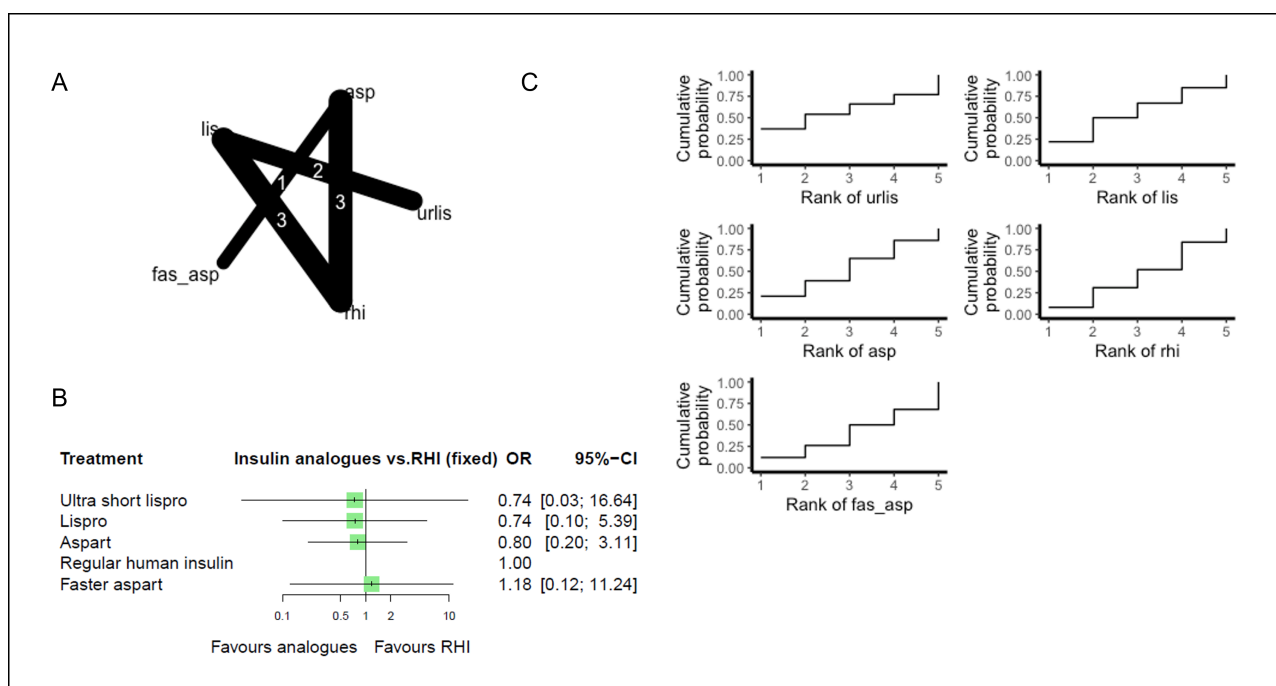


Figure 7. (A) Visual representation of the network. (B) Forest plot representing the estimates from the network meta-analysis. (C) The surface under the cumulative ranking curve (SUCRA), ranking the effectiveness of different treatments. A higher SUCRA value (from 0 to 100%) indicates a higher probability of the intervention being among the best-performing treatments. Abbreviations: lis: lispro, asp: aspart, rhi: regular human insulin, urlis: ultra rapid lispro, fas_asp: fast-acting aspart



1. Glycaemic control

See [Summary of findings 1](#); [Figure 2](#) and [Figure 3](#) (SUCRA); [Supplementary material 8](#) (league table with the effect estimates); [Supplementary material 9](#) (P scores).

1.1. Haemoglobin A1c (HbA1c)

Based on 14 trials with 6039 participants (Dreyer 2005; Home 2000; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Provenzano 2001; Raskin 2000; Recasens 2003; Z011 2007; Z013 2007; Z015 2007), all five (ultra-)short-acting insulin analogues may result in little to no difference in HbA1c compared to RHI (mean difference (MD): URLi -0.25%, 95% confidence interval (CI) -0.51 to 0.00; lispro -0.22%, 95% CI -0.46 to 0.02; glulisine -0.21%, 95% CI -0.48 to 0.06; FIAsp -0.17%, 95% CI -0.28 to -0.05; aspart -0.14%, 95% CI -0.21 to -0.06). Among the five insulins, URLi had the highest likelihood of being the most efficacious for this outcome. However, the certainty of the evidence is low for all insulin analogues due to some or major concerns about within-study bias, heterogeneity, imprecision and incoherence.

1.2. Fasting blood glucose (FBG)

Based on four trials with 2737 participants (Home 2000; Ma 2024; ONSET 1; ONSET 8), two (ultra-)short-acting insulin analogues may result in little to no difference in FBG compared to RHI, but the evidence is very uncertain (MD: FIAsp 0.69 mmol/L, 95% CI -0.31 to 1.69; aspart 0.80 mmol/L, 95% CI -0.01 to 1.61). Among the three insulins, RHI had the highest likelihood of being the most efficacious for this outcome. However, the certainty of the evidence is extremely low for both insulin analogues due to major concerns about within-study bias and unavailable assessment of imprecision, heterogeneity and incoherence because of too few trials.

2. Hypoglycaemia

See [Summary of findings 2](#); [Figure 4](#); [Figure 5](#); [Figure 6](#) (SUCRA); [Supplementary material 8](#) (league table with the effect estimates); [Supplementary material 9](#) (P scores).

a. Mild/moderate (non-severe) hypoglycaemia

Based on six trials with 3967 participants (Home 2000; Iwamoto 2001; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM), two (ultra-)short-acting insulin analogues may result in little to no difference in mild/moderate (non-severe) hypoglycaemia compared to RHI, but the evidence is very uncertain (risk ratio (RR): aspart 1.05, 95% CI 0.97 to 1.13; FIAsp 1.04, 95% CI 0.96 to 1.12). Among the three insulins, RHI had the highest likelihood of being the most efficacious for this outcome; however, the certainty of the evidence is very low for all insulin analogues due to some or major concerns about within-study bias, heterogeneity, imprecision and incoherence.

b. Severe/serious hypoglycaemia

Based on 11 trials with 5390 participants (Home 2000; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Raskin 2000; Recasens 2003; Z011 2007; Z013 2007; Z015 2007), three (ultra-)short-acting insulin analogues may result in a reduction in

severe/serious hypoglycaemia compared to RHI (RR: URLi 0.80, 95% CI 0.36 to 1.77), but the evidence is very uncertain for aspart and FIAsp (RR: aspart 0.89, 95% CI 0.73 to 1.09; FIAsp 0.79, 95% CI 0.55 to 1.15). Glulisine may result in an increase in severe/serious hypoglycaemia compared to RHI (glulisine 1.54, 95% CI 0.46 to 5.19). In contrast, lispro may result in little to no difference compared to RHI (RR: lispro 1.01, 95% CI 0.51 to 1.98). Among the five insulins, FIAsp had the highest likelihood of being the most efficacious for this outcome; however, the certainty of the evidence is low or very low due to some or major concerns about within-study bias, imprecision and incoherence.

3. Nocturnal hypoglycaemia

c. Mild/moderate nocturnal hypoglycaemia

No networks reporting this outcome were detected among the included trials (Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM).

d. Severe/serious nocturnal hypoglycaemia

Based on four trials with 3229 participants (Home 2000; ONSET 1; ONSET 8; Raskin 2000), two (ultra-)short-acting insulin analogues may result in a reduction in severe/serious nocturnal hypoglycaemia compared to RHI (RR: aspart 0.63, 95% CI 0.46 to 0.88; FIAsp 0.52, 95% CI 0.26 to 1.04), but the evidence is very uncertain. Among the three insulins, FIAsp had the highest likelihood of being the most efficacious for this outcome; however, the certainty of the evidence is very low due to some or major concerns about within-study bias, imprecision, heterogeneity and incoherence.

4. Diabetic ketoacidosis (DKA)

See [Summary of findings 3](#); [Figure 7](#) (SUCRA); [Supplementary material 8](#) (league table with the effect estimates); [Supplementary material 9](#) (P scores).

Based on nine trials with 4599 participants (Home 2000; Iwamoto 2001; Ma 2024; ONSET 1; PRONTO-T1DM; Raskin 2000; Z011 2007; Z013 2007; Z015 2007), two (ultra-)short-acting insulin analogues may result in a reduction in DKA compared to RHI (odds ratio (OR): lispro 0.74, 95% CI 0.10 to 5.39; URLi 0.74, 95% CI 0.03 to 16.64). Aspart may also result in a reduction in DKA compared to RHI (OR: aspart 0.80, 95% CI 0.20 to 3.11), but the evidence is very uncertain. On the contrary, FIAsp may result in an increase in DKA compared to RHI (OR: FIAsp 1.18, 95% CI 0.12 to 11.24). Among the four insulins, lispro had the highest likelihood of being the most efficacious for this outcome; however, the certainty of the evidence is low or very low due to some or major concerns about within-study bias, imprecision and incoherence.

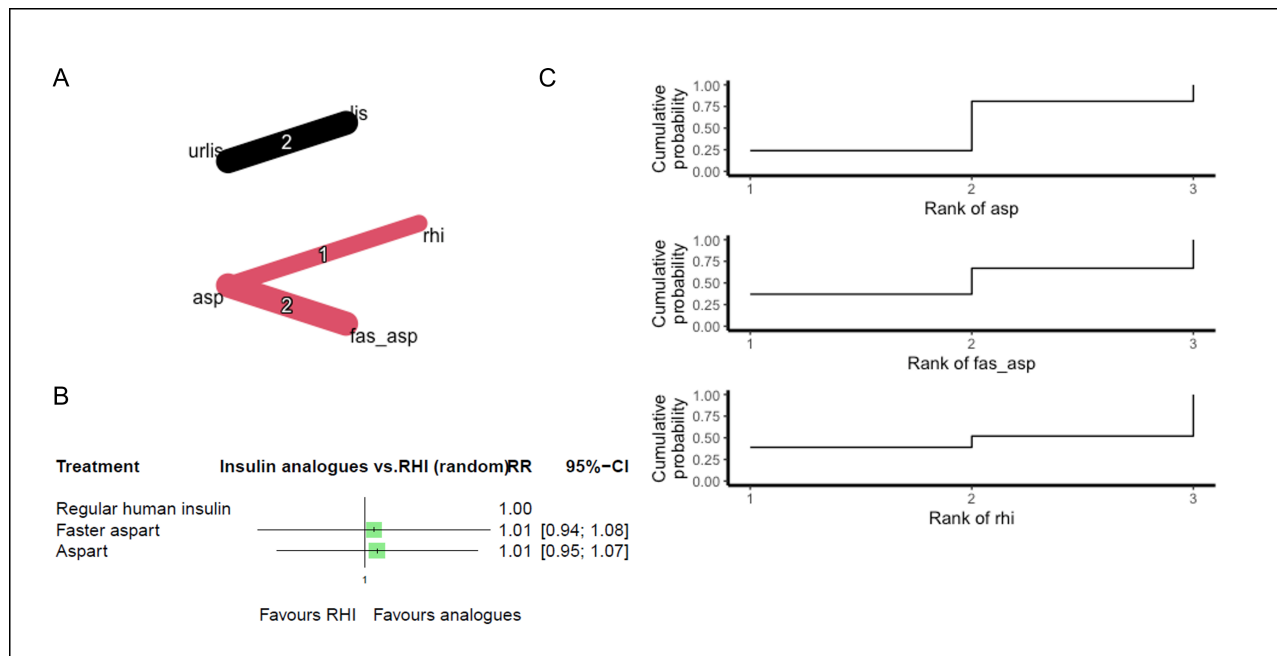
5. Quality of life

No networks reporting this outcome quantitatively were detected among the included trials (Home 2000; ONSET 1).

6. Adherence to treatment

See [Summary of findings 3](#); [Figure 8](#) (SUCRA); [Supplementary material 8](#) (league table with the effect estimates); [Supplementary material 9](#) (P scores).

Figure 8. (A) Visual representation of the network among included studies (disconnected network with regular human insulin is not further analysed). (B) Forest plot representing the estimates from the network meta-analysis. (C) The surface under the cumulative ranking curve (SUCRA), ranking the effectiveness of different treatments. A higher SUCRA value (from 0 to 100%) indicates a higher probability of the intervention being among the best-performing treatments. Abbreviations: lis: lispro, asp: aspart, rhi: regular human insulin, urlis: ultra rapid lispro, fas_asp: fast-acting aspart



Based on five trials with 3579 participants (Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Raskin 2000), two (ultra-)short-acting insulin analogues may result in little to no difference in adherence to treatment compared to RHI, but the evidence is very uncertain (RR: aspart 1.01, 95% CI 0.95 to 1.07; FIAsp 1.01, 95% CI 0.94 to 1.08). Among the three insulins, RHI had the highest likelihood of being the most efficacious for this outcome; however, the certainty of the evidence is very low due to some or major concerns about within-study bias, heterogeneity and incoherence.

Pairwise meta-analyses

Insulin lispro versus regular human insulin

Critical outcomes

1. Glycaemic control

a. HbA1c (haemoglobin A1c)

Insulin lispro results in little to no difference in HbA1c % (MD -0.22, 95% CI -0.46 to 0.02; $I^2 = 0\%$; 5 studies, 491 participants; high-certainty evidence).

b. Fasting blood glucose (FBG)

None of the included studies reported this outcome.

2. Hypoglycaemia

a. Mild/moderate (non-severe) hypoglycaemia

None of the included studies reported this outcome.

b. Severe/serious hypoglycaemia

Insulin lispro may result in little to no difference in severe/serious hypoglycaemia (Peto OR 1.00, 95% CI 0.48 to 2.09; $I^2 = 0\%$; 4 studies, 479 participants; low-certainty evidence).

3. Nocturnal hypoglycaemia

a. Mild/moderate nocturnal hypoglycaemia

None of the included studies reported this outcome.

b. Severe/serious nocturnal hypoglycaemia

None of the included studies reported this outcome.

4. Diabetic ketoacidosis (DKA)

Insulin lispro may result in little to no difference in diabetic ketoacidosis (Peto OR 0.53, 95% CI 0.05 to 5.08; $I^2 = 61\%$; 3 studies, 434 participants; low-certainty evidence).

5. Quality of life

None of the included trials reported this outcome quantitatively. The publication of Ferguson 2001 presented results on the Hypoglycaemia Fear Survey (HFS). IQWiG 2007 reported that according to the original study report, the Well Being Questionnaire (WBQ) was also used [92]. Overall, no detailed results were presented: in the publication, it was only briefly stated that there were no significant group differences, and the study report presented only information on the individual study periods, but no appropriate cross-over analysis was carried out. Z015 2007 used the Diabetes Quality of Life Clinical Trial Questionnaire (DQOLCTQ). The presented results only covered the participant populations

from the USA and Canada even though, according to the study protocol, it was planned to apply this questionnaire in European study centres [92]. Results were only presented in a qualitative manner without a description of the type of analysis carried out; there was no significant group difference.

6. Adherence to treatment

None of the included studies reported this outcome.

Important outcomes

1. Death (all-cause mortality)

The evidence is very uncertain about the effect of insulin lispro in comparison to regular human insulin on all-cause mortality (Not estimable; 4 studies, 479 participants; low-certainty evidence).

2. Macrovascular complications

a. Non-fatal myocardial infarction (MI)

None of the included studies reported this outcome.

b. Non-fatal stroke

None of the included studies reported this outcome.

3. Microvascular complications

a. Nephropathy

None of the included studies reported this outcome.

b. Retinopathy

None of the included studies reported this outcome.

4. Accidents

None of the included studies reported this outcome.

5. Adverse events

None of the included studies reported this outcome.

6. Body weight change

Insulin lispro results in little to no difference in body weight change (kg) (MD -0.60, 95% CI -1.83 to 0.64; $I^2 = 43\%$; 4 studies, 479 participants; high-certainty evidence).

Insulin aspart versus regular human insulin

Critical outcomes

1. Glycaemic control

a. HbA1c (haemoglobin A1c)

Insulin aspart likely results in little to no difference in HbA1c % (MD -0.14, 95% CI -0.21 to -0.06; $I^2 = 0\%$; 3 studies, 2143 participants; moderate-certainty evidence).

b. Fasting blood glucose (FBG)

Insulin aspart may result in little to no difference in fasting blood glucose (MD 0.80, 95% CI 0.36 to 1.24; I^2 not applicable; 1 study, 1065 participants; low-certainty evidence).

2. Hypoglycaemia

a. Mild/moderate (non-severe) hypoglycaemia

Insulin aspart may result in little to no difference in mild/moderate (non-severe) hypoglycaemia (RR 1.05, 95% CI 0.98 to 1.12; $I^2 = 0\%$; 2 studies, 1270 participants; low-certainty evidence).

b. Severe/serious hypoglycaemia

The evidence is very uncertain about the effect of insulin aspart on severe/serious hypoglycaemia (RR 0.89, 95% CI 0.73 to 1.09; $I^2 = 0\%$; 2 studies, 1947 participants; very low-certainty evidence).

3. Nocturnal hypoglycaemia

a. Mild/moderate nocturnal hypoglycaemia

None of the included studies reported this outcome.

b. Severe/serious nocturnal hypoglycaemia

Insulin aspart likely results in a reduction in severe/serious nocturnal hypoglycaemia (RR 0.63, 95% CI 0.46 to 0.88; $I^2 = 0\%$; 2 studies, 1779 participants; moderate-certainty evidence).

4. Diabetic ketoacidosis (DKA)

The evidence is very uncertain about the effect of insulin aspart on diabetic ketoacidosis (Peto OR 0.97, 95% CI 0.24 to 3.92; $I^2 = 0\%$; 3 studies, 2152 participants; very low-certainty evidence).

5. Quality of life

Insulin aspart may result in an increase in quality of life as assessed by the Diabetes Treatment Satisfaction Questionnaire (MD 2.30, 95% CI 1.32 to 3.28; I^2 not applicable; 1 study, 1065 participants; low-certainty evidence).

In the study on the insulin analogue aspart (Home 2000), results on health-related quality of life were published in Bott et al. 2003 [168], but only for the German subpopulation. The publication Home 2000 mentioned the assessment of health-related quality of life, but presented no results. IQWiG 2007 further reported that the original study report did not contain any information on measuring health-related quality of life [92], so it is unclear whether data were only assessed for the German participant subgroup or also in other study centres. According to Bott et al. 2003 [168], the Diabetes-Specific Quality of Life Scale (DSQOLS) was used. The overall questionnaire score showed no statistically significant group difference; an analysis on the subscales showed a statistically significant effect for the subscale 'diet restrictions' in favour of insulin aspart, but no statistically significant group differences for the subscales 'burden of hypoglycaemia' and 'blood glucose fluctuations'. It was unclear whether the analysis on the subscale level had been planned a priori. The baseline data presented in Bott et al. 2003 showed a difference between the two treatment groups regarding the variable gender (50% in the insulin aspart group versus 38% in the RHI group) [168], which raises concerns about whether this subgroup of participants was still appropriately randomised; the methods section provided insufficient information on this issue.

6. Adherence to treatment

Insulin aspart likely results in little to no difference in adherence to treatment (RR 1.01, 95% CI 0.97 to 1.05; I^2 not applicable; 1 study, 882 participants; moderate-certainty evidence).

Important outcomes

1. Death (all-cause mortality)

The evidence is very uncertain about the effect of insulin aspart on death (Peto OR 4.51, 95% CI 0.07 to 285.83; I^2 not applicable; 3 studies, 2152 participants; very low-certainty evidence).

2. Macrovascular complications

None of the included studies reported this outcome.

a. Non-fatal myocardial infarction (MI)

None of the included studies reported this outcome.

b. Non-fatal stroke

None of the included studies reported this outcome.

3. Microvascular complications

a. Nephropathy

None of the included studies reported this outcome.

b. Retinopathy

The evidence is very uncertain about the effect of insulin aspart on retinopathy (RR 1.31, 95% CI 0.05 to 31.78; I^2 not applicable; 1 study, 205 participants; very low-certainty evidence).

4. Accidents

None of the included studies reported this outcome.

5. Adverse events

The evidence is very uncertain about the effect of insulin aspart on mild/moderate adverse events (RR 1.02, 95% CI 0.92 to 1.14; I^2 = 30%; 2 studies, 1270 participants; very low-certainty evidence) and severe/serious adverse events (RR 0.74, 95% CI 0.45 to 1.23; I^2 = 0%; 2 studies, 1270 participants; very low-certainty evidence).

6. Body weight change

Insulin aspart likely results in little to no difference in body weight change (kg) (MD -0.10, 95% CI -0.21 to 0.01; I^2 = 23%; 2 studies, 1906 participants; moderate-certainty evidence).

Insulin glulisine versus insulin lispro

Critical outcomes

1. Glycaemic control

a. HbA1c (haemoglobin A1c)

Insulin glulisine likely results in little to no difference in HbA1c (MD 0.01, 95% CI -0.11 to 0.13; I^2 = 0%; 2 studies, 920 participants; moderate-certainty evidence).

b. Fasting blood glucose (FBG)

None of the included studies reported this outcome.

2. Hypoglycaemia

a. Mild/moderate (non-severe) hypoglycaemia

None of the included studies reported this outcome.

b. Severe/serious hypoglycaemia

The evidence is very uncertain about the effect of insulin glulisine on severe/serious hypoglycaemia (RR 1.53, 95% CI 0.56 to 4.19; I^2 not applicable; 1 study, 267 participants; very low-certainty evidence).

3. Nocturnal hypoglycaemia

a. Mild/moderate nocturnal hypoglycaemia

None of the included studies reported this outcome.

b. Severe/serious nocturnal hypoglycaemia

None of the included studies reported this outcome.

4. Diabetic ketoacidosis (DKA)

None of the included studies reported this outcome.

5. Quality of life

None of the included studies reported this outcome.

6. Adherence to treatment

None of the included studies reported this outcome.

Important outcomes

1. Death (all-cause mortality)

None of the included studies reported this outcome.

2. Macrovascular complications

None of the included studies reported this outcome.

a. Non-fatal myocardial infarction (MI)

None of the included studies reported this outcome.

b. Non-fatal stroke

None of the included studies reported this outcome.

3. Microvascular complications

None of the included studies reported this outcome.

a. Nephropathy

None of the included studies reported this outcome.

b. Retinopathy

None of the included studies reported this outcome.

4. Accidents

None of the included studies reported this outcome.

5. Adverse events

The evidence is very uncertain about the effect of insulin glulisine on mild/moderate adverse events (RR 0.99, 95% CI 0.89 to 1.10; I^2 = 33%; 2 studies, 939 participants; very low-certainty evidence) and severe/serious adverse events (RR 1.29, 95% CI 0.91 to 1.81; I^2 = 0%; 2 studies, 939 participants; very low-certainty evidence).

6. Body weight change

None of the included studies reported this outcome.

Ultra-rapid lispro insulin versus insulin lispro

Critical outcomes

1. Glycaemic control

a. HbA1c (haemoglobin A1c)

Ultra-rapid lispro insulin results in little to no difference in HbA1c % (MD -0.02, 95% CI -0.15 to 0.11; $I^2 = 45\%$; 2 studies, 1136 participants; high-certainty evidence).

b. Fasting blood glucose (FBG)

Ultra-rapid lispro insulin likely results in little to no difference in fasting blood glucose (MD -0.45, 95% CI -1.00 to 0.10; I^2 not applicable; 1 study, 329 participants; moderate-certainty evidence).

2. Hypoglycaemia

a. Mild/moderate (non-severe) hypoglycaemia

Ultra-rapid lispro insulin results in little to no difference in mild/moderate (non-severe) hypoglycaemia (RR 0.99, 95% CI 0.96 to 1.03; $I^2 = 0\%$; 2 studies, 1247 participants; high-certainty evidence).

b. Severe/serious hypoglycaemia

Ultra rapid lispro insulin may result in a reduction in severe/serious hypoglycaemia (RR 0.79, 95% CI 0.52 to 1.21; $I^2 = 0\%$; 2 studies, 1247 participants; low-certainty evidence).

3. Nocturnal hypoglycaemia

a. Mild/moderate nocturnal hypoglycaemia

Ultra-rapid lispro insulin likely results in little to no difference in mild/moderate nocturnal hypoglycaemia (RR 0.97, 95% CI 0.86 to 1.11; $I^2 = 0\%$; 2 studies, 1247 participants; moderate-certainty evidence).

b. Severe/serious nocturnal hypoglycaemia

None of the included studies reported this outcome.

4. Diabetic ketoacidosis (DKA)

Ultra-rapid lispro insulin may result in little to no difference in diabetic ketoacidosis (Peto OR 1.00, 95% CI 0.06 to 15.92; $I^2 = 51\%$; 2 studies, 1247 participants; low-certainty evidence).

5. Quality of life

None of the included studies reported this outcome.

6. Adherence to treatment

Ultra-rapid lispro insulin results in little to no difference in adherence to treatment (RR 0.99, 95% CI 0.95 to 1.03; $I^2 = 51\%$; 2 studies, 1247 participants; high-certainty evidence).

Important outcomes

1. Death (all-cause mortality)

Ultra-rapid lispro insulin may result in little to no difference in all-cause mortality (Peto OR 0.98, 95% CI 0.06 to 15.69; I^2 not applicable; 2 studies, 1247 participants; low-certainty evidence).

2. Macrovascular complications

a. Non-fatal myocardial infarction (MI)

Ultra-rapid lispro insulin may result in little to no difference in non-fatal myocardial infarction (Peto OR 1.00, 95% CI 0.06 to 15.92; $I^2 = 51\%$; 2 studies, 1247 participants; low-certainty evidence).

b. Non-fatal stroke

Ultra-rapid lispro insulin may result in little to no difference in non-fatal stroke (Peto OR 1.00, 95% CI 0.06 to 15.92; $I^2 = 49\%$; 2 studies, 1247 participants; low-certainty evidence).

3. Microvascular complications

a. Nephropathy

Ultra-rapid lispro insulin may result in little to no difference in nephropathy (Peto OR 7.47, 95% CI 0.15 to 376.66; I^2 not applicable; 1 study, 354 participants; low-certainty evidence).

b. Retinopathy

None of the included studies reported this outcome.

4. Accidents

None of the included studies reported this outcome.

5. Adverse events

Ultra-rapid lispro insulin likely results in little to no difference in mild/moderate adverse events (RR 1.05, 95% CI 0.92 to 1.19; $I^2 = 0\%$; 2 studies, 1247 participants; moderate-certainty evidence), but it may result in a reduction in severe/serious adverse events (RR 0.83, 95% CI 0.60 to 1.14; $I^2 = 0\%$; 2 studies, 1247 participants; low-certainty evidence).

6. Body weight change

None of the included studies reported this outcome.

Fasting-acting aspart insulin versus insulin aspart

Critical outcomes

1. Glycaemic control

a. HbA1c (haemoglobin A1c)

Fast-acting insulin aspart results in little to no difference in HbA1c (MD -0.03, 95% CI -0.13 to 0.07; $I^2 = 22\%$; 2 studies, 1349 participants; high-certainty evidence).

b. Fasting blood glucose (FBG)

Fast-acting insulin aspart may result in little to no difference in fasting blood glucose (MD -0.11, 95% CI -0.70 to 0.48; $I^2 = 68\%$; 2 studies, 1343 participants; low-certainty evidence).

2. Hypoglycaemia

a. Mild/moderate (non-severe) hypoglycaemia

Fast-acting insulin aspart results in little to no difference in mild/moderate (non-severe) hypoglycaemia (RR 0.99, 95% CI 0.97 to 1.02; $I^2 = 63\%$; 2 studies, 1450 participants; high-certainty evidence).

b. Severe/serious hypoglycaemia

Fast-acting insulin aspart may result in a reduction in severe/serious hypoglycaemia (RR 0.89, 95% CI 0.65 to 1.21; $I^2 = 0\%$; 2 studies, 1450 participants; low-certainty evidence).

3. Nocturnal hypoglycaemia

a. Mild/moderate nocturnal hypoglycaemia

Fast-acting insulin aspart likely results in little to no difference in mild/moderate nocturnal hypoglycaemia (RR 0.99, 95% CI 0.92 to 1.06; $I^2 = 23\%$; 2 studies, 1450 participants; moderate-certainty evidence).

b. Severe/serious nocturnal hypoglycaemia

Fast-acting insulin aspart may result in little to no difference in severe/serious nocturnal hypoglycaemia (RR 0.83, 95% CI 0.45 to 1.51; $I^2 = 0\%$; 2 studies, 1450 participants; low-certainty evidence).

4. Diabetic ketoacidosis (DKA)

Fast-acting insulin aspart likely results in little to no difference in diabetic ketoacidosis (OR 1.48, 95% CI 0.25 to 8.91; I^2 not applicable; 1 study, 766 participants; moderate-certainty evidence).

5. Quality of life

Fast-acting insulin aspart likely results in little to no difference in quality of life as measured by the SF-36 overall score for physical health (higher is better) (MD -0.99, 95% CI -2.12 to 0.14; I^2 not applicable; 1 study, 695 participants; moderate-certainty evidence) and the SF-36 overall score for mental health (higher is better) (MD -0.46, 95% CI -1.90 to 0.98; I^2 not applicable; 1 study, 695 participants; moderate-certainty evidence).

6. Adherence to treatment

Fast-acting insulin aspart results in little to no difference in adherence to treatment (RR 1.00, 95% CI 0.97 to 1.03; $I^2 = 75\%$; 2 studies, 1450 participants; high-certainty evidence).

Important outcomes

1. Death (all-cause mortality)

Fast-acting insulin aspart may result in little to no difference in all-cause mortality (Peto OR 0.13, 95% CI 0.00 to 6.71; I^2 not applicable; 2 studies, 1450 participants; low-certainty evidence).

2. Macrovascular complications

a. Non-fatal myocardial infarction (MI)

Fast-acting insulin aspart may result in little to no difference in non-fatal myocardial infarction (Peto OR 7.39, 95% CI 0.15 to 372.38; I^2 not applicable; 2 studies, 1450 participants; low-certainty evidence).

b. Non-fatal stroke

Fast-acting insulin aspart may result in little to no difference in non-fatal stroke (Peto OR 0.13, 95% CI 0.00 to 6.71; I^2 not applicable; 2 studies, 1450 participants; low-certainty evidence).

3. Microvascular complications

a. Nephropathy

Fast-acting insulin aspart results in little to no difference in nephropathy (Peto OR 0.24, 95% CI 0.05 to 1.04; $I^2 = 0\%$; 2 studies, 1450 participants; high-certainty evidence).

b. Retinopathy

Fast-acting insulin aspart likely results in little to no difference in retinopathy (RR 1.44, 95% CI 0.71 to 2.91; $I^2 = 0\%$; 2 studies, 1450 participants; moderate-certainty evidence).

4. Accidents

Fast-acting insulin aspart may result in little to no difference in accidents (RR 1.04, 95% CI 0.83 to 1.32; $I^2 = 0\%$; 2 studies, 1450 participants; low-certainty evidence).

5. Adverse events

Fast-acting insulin aspart results in little to no difference in mild/moderate adverse events (RR 0.99, 95% CI 0.94 to 1.04; $I^2 = 0\%$; 2 studies, 1450 participants; high-certainty evidence). In addition, it may result in little to no difference in severe/serious adverse events (RR 1.09, 95% CI 0.75 to 1.57; $I^2 = 0\%$; 2 studies, 1450 participants; low-certainty evidence).

6. Body weight change

Fast-acting insulin aspart results in little to no difference in body weight change (kg) (MD -0.07, 95% CI -1.16 to 1.02; $I^2 = 32\%$; 2 studies, 1366 participants; high-certainty evidence).

Subgroup analyses

We could not perform most of the planned subgroup analyses because the available data were insufficient. For six critical outcomes for which we carried out pairwise meta-analyses, we performed network meta-analyses for (ultra-)short-acting insulin analogues and RHI. The results are presented in the sections on the respective outcomes.

Sensitivity analyses

Due to the few included trials, we were unable to perform sensitivity analysis for either pairwise or network analyses for trials with low overall risk of bias only.

Equity assessment

Distribution of effects across PROGRESS-PLUS characteristics

We could not perform analyses restricted to population, intervention and setting (see above 'Subgroup analyses'). All included trials stated trial locations and gender percentages. Eight trials described information about ethnicity/race (Home 2000; Iwamoto 2001; Kawamori 2009; Ma 2024; ONSET 1; ONSET 8; PRONTO-T1DM; Raskin 2000), but participants were mainly White or Asian populations. No studies elaborated on elder groups (over 65 years), socioeconomic status, residential places, or educational or occupational status. Healthcare settings were presumably outpatient, but no further descriptions could be obtained.

Studies focusing on disadvantaged or at-risk populations

None of the trials in this review focused on minority groups or people in low-income countries (LICs). Five trials were launched in multinational centres including several middle-income countries (MICs) (Dreyer 2005; Ma 2024; ONSET 8; Z011 2007; Z013 2007). Only Ma 2024 was exclusively launched in one MIC. These five trials developed tailored interventions targeting these groups.

Reporting biases

Due to the limited number of included studies in the pairwise comparisons with RHI, we could not draw funnel plots for any critical or important outcomes. For the network meta-analysis, we could draw funnel plots for two critical outcomes (each with more than 10 studies), HbA1c (Figure 2 (B)) and severe/serious hypoglycaemia (Figure 5 (B)), which had at least 10 studies. We did not identify evidence of publication bias in these comparisons.

DISCUSSION

Summary of main results

We included 15 RCTs with 6335 randomised participants, assessing the treatment effects of five (ultra-)short-acting insulin analogues (URLi, lispro, glulisine, FIAsp and aspart) compared to regular human insulin (RHI). The main findings of the network meta-analysis for the critical outcomes are as follows (Summary of findings 1; Summary of findings 2; Summary of findings 3):

All five (ultra-)short-acting insulin analogues may result in little to no difference in HbA1c compared to RHI. For fasting blood glucose, we are uncertain about the effects of aspart and FIAsp. Lispro, glulisine and URLi could not be assessed in relation to RHI. We are uncertain about the effects of aspart and FIAsp in mild to moderate (non-severe) hypoglycaemia, whereas lispro, glulisine and URLi could not be assessed in relation to RHI. For severe/serious hypoglycaemia, URLi may result in a reduction compared to RHI, but the evidence is very uncertain for aspart and FIAsp. Glulisine may result in an increase in severe/serious hypoglycaemia compared to RHI, whereas lispro may result in little to no difference compared to RHI. We are also uncertain about the effects of aspart and FIAsp in severe/serious nocturnal hypoglycaemia, whereas lispro, glulisine and URLi could not be assessed in relation to RHI. Furthermore, lispro and URLi may result in a reduction in diabetic ketoacidosis (DKA) compared to RHI, but the evidence is very uncertain for aspart. FIAsp may result in an increase in DKA compared to RHI, whereas glulisine could not be assessed in relation to RHI. For treatment adherence, we are uncertain about the effects of aspart and FIAsp, whereas lispro, glulisine and URLi could not be assessed in relation to RHI. Additionally, we were unable to evaluate mild to moderate nocturnal hypoglycaemia and quality of life using a network meta-analysis.

Limitations of the evidence included in the review

Eleven of the included studies were conducted unblinded and lacked study protocols or analysis details, raising concerns about biases in randomisation, deviations in intervention delivery and selective reporting. Four studies on ultra-short-acting insulins had a relatively lower risk of bias in those domains. All trials had low or some concerns about risk of bias due to missing outcomes, primarily resulting from low or moderate attrition rates. We therefore detected different overall risks of bias among the included studies.

There were several inconsistencies in the reporting of methods and results across studies on short-acting insulin analogues. Both studies on insulin aspart showed inconsistencies in the reporting of the results on HbA1c and severe hypoglycaemic events (Home 2000; Raskin 2000). In Home 2000, the reported HbA1c value in the text of the publication differed from the value presented in the table. This discrepancy was likely to be due to one of the

values being adjusted for other baseline variables; however, there was no clear explanation. For Raskin 2000, a different baseline HbA1c value was reported in Lindholm 2002 [169], a review article that was published by one of the co-authors of the original trial. Furthermore, the definition of hypoglycaemia and also the time frame used to define nocturnal hypoglycaemia showed inconsistencies: in the publication of Raskin 2000, hypoglycaemic events were divided up into minor (blood glucose value less than 45 mg/dL (2.5 mmol/L) or classical symptoms of hypoglycaemia) and major events (need of assistance or administration of parenteral glucose or glucagon), while the original study report described a further distinction of major events into grade A and B. Both Home 2000 and Raskin 2000 defined nocturnal hypoglycaemia as a hypoglycaemic episode that occurred between 0:00 and 8:00 am in the original study report, but between 0:00 and 6:00 am in the publications [92]. The same discrepancy regarding the definition of nocturnal hypoglycaemia occurred with Ferguson 2001. There were inconsistent results regarding severe hypoglycaemia in the study report of Z11 2007 and the review by Brunelle et al. [7], a co-author of the original trial. Among trials on ultra-short-acting insulin analogues (URAs), severe hypoglycaemia was defined as < 3.9 mmol/L (70 mg/dL) in Ma 2024, whereas ONSET 1, ONSET 8 and PRONTO-T1DM used < 3.1 mmol/L (56 mg/dL).

A commonly used criterion for the definition of severe hypoglycaemic episodes was the need for assistance from a third person. However, this type of definition is highly subjective and therefore prone to bias. A more robust definition, such as 'injection of glucose or glucagon by another person', may have resulted in more reliable data [170].

Caution is also necessary when interpreting the results regarding health-related quality of life. For insulin aspart, results were only presented in the publication by Bott et al. [168], while the outcome was not mentioned in the original study report. The publication for Home 2000 reported that health-related quality of life was assessed, but did not report any results (see IQWiG 2007 [92]). Bott et al. only reported results for the German subpopulation [168]. Overall, it was unclear whether this was a retrospective analysis. Differences in the baseline data of the two treatment groups in Bott et al. raised the question of whether the groups were adequately randomised [168].

As previously, we included only studies with a follow-up duration of at least 24 weeks. This restriction was intended to focus better on the effects of insulin analogues on participant-relevant outcomes. To draw definitive conclusions about long-term outcomes, such as mortality or microvascular and macrovascular complications of diabetes, trials with a follow-up duration of several years would be required. The longest trials found in our systematic search had a follow-up duration of 12 months; therefore, we could not provide any evidence for long-term follow-up.

Only Home 2000 and four trials on URAs reported fasting glucose values, but the measurement in Home 2000 was from self-monitoring. From the publication, we observed clear differences among the pre-breakfast, pre-lunch, and pre-dinner self-monitoring measurements. Due to a lack of fasting plasma measurement, we retrieved the one prior to breakfast. This may explain the substantial statistical heterogeneity in our network meta-analysis among the three connected trials (Home 2000; ONSET 1; ONSET 8). However, we were unable to confirm this with subgroup analyses.

The results of this review may not be generalised to all people with type 1 diabetes due to intervention routes, timing and follow-up durations. Many trials had restrictive participant selection criteria, excluding participants with diabetic complications. There were also no trials in this review in which participants used a continuous subcutaneous insulin infusion (CSII) delivery system. However, URAAs have been recommended for patients with type 1 diabetes using CSII in several countries, such as the UK, Canada and the USA [16, 171, 172, 173]. Despite its importance [174], we were unable to inspect the optimal timing of those injections, given that only ONSET 8 provided data at mealtime and post-mealtime. This will be covered in another Cochrane review on insulin regimens that our team is preparing for submission (PROSPERO registration: CRD 420251163760). As it has been shown that insulin pumps might be associated with favourable long-term outcomes and are increasingly used by participants, this is a major gap in the evidence on (ultra-)short-acting insulin analogues in this review [16, 175, 176, 177, 178, 179]. However, there are trials with follow-up periods shorter than 24 weeks that show a beneficial effect of (ultra-)short-acting insulin analogues on glycaemic control compared to RHI [178, 180, 181, 182, 183].

It also should be noted that the majority of participants treated with short-acting insulin analogues used NPH as basal insulin, whereas the ones treated with URAAs used various long-acting insulins such as degludec, glargine, detemir or metformin (Ma 2024). Long-acting insulins were administered once or twice daily. Different ultra-long-/long-/intermediate-acting insulin could have an influential impact on identical short-term and long-term diabetes-related outcomes to (ultra-)short-acting insulin analogues [184]. Therefore, little can be concluded about whether the observed effects would be the same if a long-acting insulin analogue was used instead.

Last, but not least, this review did not take modern technologies into consideration, such as continuous glucose monitoring (CGM). The National Institute for Health and Care Excellence (NICE) in England has recommended that two categories of CGM, either real-time CGM or intermittent CGM, are offered for adults with type 1 diabetes, provided that equitable access and care support is provided [185]. Concordant with NICE, the American Diabetes Association underscored the essential use of CGM as standard of care for type 1 diabetes, but recommended self-monitoring of glucose measurement as a long-lasting backup [49]. In this regard, time-in-range (TIR) might add diagnostic and prognostic value in glycaemic control apart from HbA1c [186], such as during pregnancy [49]. However, debates about the definitions and explicit roles of TIR exist [187, 188].

Limitations of the review processes

Overall, our results were based on trials identified through an extensive and systematic literature search, including articles in all languages. We also searched trial registers to identify potentially relevant trials that have not yet been published. However, due to our restrictions of only including longer trials and RCTs, the number of included trials is very low and, therefore, the insights gained on many outcomes are mostly inconclusive. On the one hand, this clearly indicates the lack of firm evidence available regarding many outcomes; on the other hand, numerous shorter trials or trials using observational designs are often overlooked. To gain a better understanding of the value of short-acting insulin analogues in general, all available evidence should be considered.

Furthermore, this review focusses only on a narrow question, comparing (ultra-)short-acting insulin analogues to RHI, with all other diabetes-related medications being the same in both groups. While this allows us to single out the effect that can be obtained by the use of short-acting insulin analogues alone, some effects might only become evident if full analogue insulin regimens (combining short-acting with basal insulins) or other supportive manoeuvres, such as oral antidiabetic medications, frequent professional counselling, and dietary and physical activity counselling [172, 189], are compared to RHI-only regimens [190, 191, 192, 193].

Our review process was potentially hampered by insufficient and inconsistent information provided in the available publications. Although we, and previous review authors, contacted nearly all trial authors, we obtained no further details to complete the missing information. The lack of information regarding study design and applied methods made it difficult to judge the risk of bias in many cases, so that we frequently had to judge the risk of bias as unclear.

For severe hypoglycaemia, we combined data from several trials in a meta-analysis. However, it is important to keep in mind that the definitions of severe hypoglycaemic episodes, as well as the trial population, varied across trials. For example, the majority of the trials excluded people with hypoglycaemia unawareness (Home 2000; Iwamoto 2001; Ma 2024; ONSET 1; PRONTO-T1DM; Provenzano 2001; Raskin 2000; Z11 2007; Z13 2007; Z15 2007), while one trial explicitly included only these participants (Ferguson 2001). Potentially, a beneficial effect of insulin analogues on avoiding severe hypoglycaemic effects would become more apparent in participants at particular risk of hypoglycaemia. However, since most trials excluded these participants, we lacked sufficient data to investigate this question.

Furthermore, as mentioned above, due to the lack of blinding and subjective definitions of hypoglycaemia, the results were at a high risk of performance bias. The data on severe hypoglycaemia reported in the included trials were also not detailed enough to carry out a more thorough meta-analysis. We only looked at the number of participants who experienced at least one severe hypoglycaemic episode, but did not take into account the total or average number of episodes experienced in the two groups.

Finally, this review does not provide any information regarding the use of insulin analogues in children or pregnant women, as we explicitly excluded these groups.

Agreements and disagreements with other studies or reviews

Our finding of no clinical benefits in glycaemic control with (ultra-)short-acting insulin analogues over RHI differs from the previous systematic review [45] and the results of other published literature reviews [19, 32, 194, 195, 196, 197]. A meta-analysis including 27 studies of diabetes found that the three short-acting insulins, aspart, lispro and glulisine, compared to RHI, were more effective at reducing postprandial glucose (PPG) and HbA1c in type 1 diabetes, but the evidence had high heterogeneity and did not differentiate these insulin analogues [198]. A recent review found that URLi had faster absorption and shorter insulin action, but it focused on PPG and CSII users [199]. Concordant with this finding, Stamati et al. reported that URAAs were better than short-acting

insulins for controlling PPG and hypoglycaemia among CSII users [178].

We also reported no differences in treatment adherence among these insulin types as compared to RHI. However, adherence to insulin treatment in all types of diabetes seems to be a longstanding issue because of insufficient qualified measurements [200] and multiple intervention-related barriers, such as the use of injections for insulin delivery and the fear of hypoglycaemia episodes [201]. Undoubtedly, similar to children and adolescents [202], treatment adherence represents a cornerstone for long-term effects in adults with type 1 diabetes.

Regarding hypoglycaemia, our findings agreed with other reviews that did not find substantial differences between insulin analogues and RHI regarding non-severe hypoglycaemic episodes [194, 195, 203]. Several reviews which, in contrast to ours, also included studies of shorter duration, reported a reduction of nocturnal hypoglycaemia under the treatment with insulin analogues [178, 194, 195, 203]. Due to the low number of studies investigating non-severe nocturnal hypoglycaemia in our review, as well as the inconsistencies in reporting on the definition of nocturnal hypoglycaemia, we did not carry out a network meta-analysis on this outcome and suggest interpreting the results with caution. However, we found that ultra-short-acting insulin analogues may have clinical benefits regarding the occurrence of severe hypoglycaemia and diabetic ketoacidosis, which have been unclear from previous reviews [197, 198]. Although Melo et al. reported a decreased risk of severe hypoglycaemia episodes by using short-acting insulins, no further information was provided, and the follow-ups were mainly shorter than six months [196].

Regarding health-related quality of life, studies with shorter follow-up not included in this review suggest improved outcomes with insulin analogues compared to RHI [182, 196, 204, 205]. However, this result is almost always related to insulin analogues being perceived as more convenient, as they can be administered with meals rather than 20 to 30 minutes beforehand. As described above, it is unclear whether this insulin-meal interval is truly necessary for RHI.

Overall, there was a lack of observational studies reporting on the long-term benefits of insulin analogues for durations exceeding one year and the harms of short-term insulin analogues for durations less than six months.

AUTHORS' CONCLUSIONS

Implications for practice

For adults with type 1 diabetes using multiple daily injections, our analysis shows that (ultra-)short-acting insulin analogues may have clinical benefits with regard to severe/serious hypoglycaemia and diabetic ketoacidosis compared to regular human insulin at short-term follow-up. They may make little to no difference to glycaemic control and mild to moderate hypoglycaemia. The evidence remains unclear regarding mild to moderate nocturnal hypoglycaemia and their impact on quality of life. There were no data for long-term follow-up. The certainty of the evidence is limited by the small number of studies, insufficient data for analysis and inconsistent definitions or measurements of some outcomes.

Equity-related implications for practice

Although we conclude that (ultra-)short-acting insulin analogues may be equivalent to regular human insulin regarding glycaemic control and non-severe hypoglycaemia, they may reduce the incidence of severe hypoglycaemia and diabetic ketoacidosis in adults with type 1 diabetes. However, these conclusions are not sufficiently robust for generalisation.

First, our included trials provided a narrow focus in terms of participants with type 1 diabetes on multiple daily injections. We investigated comparisons exclusively through randomised controlled trials in adults, but not in pregnant women. None of the included trials were conducted in low-income countries. Furthermore, in terms of the health equity-related characteristics defined by the PROGRESS-Plus framework, we could not obtain complete information on trial location, race/ethnicity or gender across the trials. No trials reported on any low-resource settings or enrolled marginalised groups. In addition, the evidence might not be generalisable to the elderly group (over 65 years).

Second, the availability and affordability of insulin analogues have been questioned most prominently in low- and middle-income countries. In some settings, only biosimilar forms of (ultra-)short-acting insulins may be available [206], which were excluded in our comparisons. This also applies to long-acting insulin analogues, essential in basal-bolus regimens, which were not assessed in our review.

Third, the application environment of these insulins was presumably a community or primary care setting. In-hospital or emergency care situations might need to be cautiously conceptualised based on individual considerations [207, 208].

Implications for research

To assess safety, high-quality studies with large numbers of participants using short-acting insulin analogues and long-term follow-up are needed. There is insufficient information on the development of long-term adverse events. Furthermore, there is a need for further research on the effects of short-acting insulin analogues on nighttime non-severe hypoglycaemia, the development of long-term complications and health-related quality of life, as well as the cost-effectiveness of this treatment.

Future research will need to take into account new, even faster-acting insulins currently under development and being tested [209, 210, 211]. In addition, methods of insulin application will likely become more variable in the future. People are increasingly using various types of insulin pumps, and new research shows that modulation of the injection site or other needle-free applications can affect the pharmacokinetic and pharmacodynamic profiles of short-acting insulins [212, 213, 214].

Currently, a comprehensive model of care is strongly recommended for type 1 diabetes. Both short- and long-acting insulin types have significant clinical implications for clinical outcomes, and the evaluation of their combination in regimens is also important to explore [215]. Dietary interventions and physical activity or other medications could also have a considerable impact [172]. Therefore, a novel method to assess the evidence on the combined effects of all glucose control management interventions could be warranted.

Equity-related implications for research

Future research on (ultra-)short-acting insulin analogues should give greater consideration to equity-related characteristics. This might not be restricted to enrolling more diverse groups of participants, such as marginalised groups and the elderly, but could also emphasise the possible settings for self-management, such as home or community care. Trial designs based on the PROGRESS-Plus framework could significantly influence the interpretation of the evidence.

Methodologically, the need to evaluate the role of these insulins in the context of combined interventions for people with type 1 diabetes cannot be ignored. They often need a holistic treatment plan that encompasses basal insulin therapy, dietary counselling, physical activity, and support to enhance adherence and reduce treatment burden [172].

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD012161.pub2](https://doi.org/10.1002/14651858.CD012161.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Threshold for minimally important differences used for GRADE judgements for the outcomes of the review

Supplementary material 8 League table for critical outcomes: HbA1c, fasting blood glucose (FBG), mild/moderate (non-severe) hypoglycaemia, severe/serious hypoglycaemia, severe/serious nocturnal hypoglycaemia, diabetic ketoacidosis(DKA), adherence to treatment

Supplementary material 9 P-scores for critical outcomes: HbA1c, fasting blood glucose (FBG), mild/moderate (non-severe) hypoglycaemia, severe/serious hypoglycaemia, severe/serious nocturnal hypoglycaemia, diabetic ketoacidosis(DKA), adherence to treatment

Supplementary material 10 Conflicts of interest declared by study investigators/authors in included studies

Supplementary material 11 Summary of findings table - Insulin lispro compared to regular human insulin for adults with type 1 diabetes mellitus on multiple daily injections

Supplementary material 12 Summary of findings table - Insulin aspart compared to regular human insulin for adults with type 1 diabetes mellitus on multiple daily injections

Supplementary material 13 Summary of findings table - Insulin glulisine compared to insulin lispro for adults with type 1 diabetes mellitus on multiple daily injections

Supplementary material 14 Summary of findings table - Ultra rapid lispro insulin compared to insulin lispro for adults with type 1 diabetes mellitus on multiple daily injections

Supplementary material 15 Summary of findings table - Fast-acting insulin aspart compared to insulin aspart for adults with type 1 diabetes mellitus on multiple daily injections

ADDITIONAL INFORMATION

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We did not use any artificial intelligence tools for the conduct of this review, except for software to check the accuracy of the language.

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- Sign-off Editor (final editorial decision): Jamie Hartmann-Boyce, University of Massachusetts Amherst;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Luisa Fernandez Mauleffinch, Cochrane Central Editorial Service;
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Yang Guo: methodology, writing, review and editing.

Yuqian Mei: methodology, writing, review.

Brenda Bongaerts: methodology, screening, writing and editing of the review.

Nandi Siegfried: feedback on GRADE methodology.

Maria-Inti Metzendorf: information specialist, writing.

Birgit Fullerton, Andrea Siebenhofer, Klaus Jeitler, Karl Horvath, Thomas Semlitsch, Andrea Berghold and Johannes Plank contributed to the development of the previous version of the review and reviewed and approved the current update.

Juan Franco: conceptualisation, methodology, writing, review and editing. He is the guarantor of the review.

All authors approved the final version for publication and agree to be accountable for all aspects of the review.

Declarations of interest

YG: none known.

YQM: none known.

BB: none known; Senior Editor for the Cochrane Database of Systematic Reviews and Coordinating Editor for Cochrane Metabolic and Endocrine Disorders, but she was not involved in the editorial process for this update.

MIM: none known; MIM is a Cochrane Editor but was not involved in the editorial process for this update.

BF: none known.

AS: was involved in the preparation of the report on short-acting insulin analogues for the treatment of type 1 diabetes mellitus for the Institute for Quality and Efficiency in Health Care (www.iqwig.de).

KJ: was involved in the preparation of the report on short-acting insulin analogues for the treatment of type 1 diabetes mellitus for the Institute for Quality and Efficiency in Health Care (www.iqwig.de).

KH: was involved in the preparation of the report on short-acting insulin analogues for the treatment of type 1 diabetes mellitus for the Institute for Quality and Efficiency in Health Care (www.iqwig.de). KH has received payment for lectures, travel/accommodations/meeting expenses and consultancy from various sources (Novartis, Medtronic, Eli Lilly, Novo Nordisk, Sanofi Aventis, Merck Sharp & Dohme).

TS: none known.

AB: none known.

JP: none known.

What's new

Date	Event	Description
19 June 2026	New search has been performed	Search updated 14 May 2025. We updated the inclusion criteria to include ultra-rapid-acting insulins and network meta-analysis. We excluded trials related to insulin pump use.
19 June 2026	New citation required and conclusions have changed	We had to expand the search strategy to cover the new drugs, and we aligned the findings of this review to the upcoming WHO Guideline on adults with type 1 diabetes mellitus on multiple daily injections. We prioritised a broad set of critical and impor-

NS: methodologist in the World Health Organization for guidelines on insulin management in type 1 diabetes mellitus.

JF: no relevant interests; Senior Editor for the Cochrane Database of Systematic Reviews and Managing Editor for Cochrane Metabolic and Endocrine Disorders, but was not involved in the editorial process for this update.

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Internal sources

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Provides in-kind support to JVAF and BB for the conduct of Cochrane reviews.

External sources

- World Health Organization, Switzerland

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Registration and protocol

The protocol for this review was prospectively registered in PROSPERO (CRD42024599817) on 1 November 2024. This is an update of the previous review [45], which was first published in 2006 [216]. The methods have been adapted and updated to current standards.

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library: Full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies included ([Supplementary material 2](#)), ongoing or awaiting classification, or excluded at the full-text screen, in the final review ([Supplementary material 3](#); [Supplementary material 4](#)); study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 5](#); [Supplementary material 6](#)). Appropriate permissions have been obtained for such use. Analyses and data management were conducted using Cochrane's authoring tool, RevMan, using built-in computation methods.

Date	Event	Description
		tant outcomes. We incorporated six new trials with 3642 participants, providing an extensive evaluation of insulin analogues in comparison to regular human insulin or head-to-head comparisons, with the incorporation of network meta-analysis, RoB 2 assessment and CINEMA.

History

Review first published: Issue 6, 2016

Date	Event	Description
27 June 2019	Amended	Conflict of interest statement in published Cochrane review and Conflict of Interest form were harmonised.
29 February 2016	New citation required but conclusions have not changed	The conclusion drawn from the first update of the original systematic review remained unchanged.
29 February 2016	New search has been performed	This review is an update of the former Cochrane review "Short acting insulin analogues versus regular human insulin in patients with diabetes mellitus", which has been withdrawn and split into two Cochrane reviews on short-acting insulin analogues versus regular human insulin for type 1 and 2 diabetes mellitus.

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ADDITIONAL TABLES

Table 1. Overview of studies included in synthesis (study characteristics, intervention and comparisons)

Study ID	Country	Setting	Study period	Comparison	Number of participants randomised	Duration of treatment (duration of follow-up)	Synthesis
Short-acting insulins (11 studies)							
Dreyer 2005	13 European countries and South Africa	Multinational, multi-centre	-	I: glulisine	342	26 weeks	Yes
				C: lispro	341		
Ferguson 2001	Scotland	Outpatient clinic of the Department of Diabetes, Royal Infirmary of Edinburgh	-	I: lispro (premeal)	39	24 weeks ^a (12 months)	No
				C: RHI (premeal)			
Home 2000	Austria, Denmark, Finland, Germany, Norway, Sweden, Switzerland, UK	Multicentre, outpatients	1997	I: aspart (premeal)	708	6 months	Yes
				C: RHI (premeal)	362		
Iwamoto 2001	Japan	Hospital outpatients; physicians' clinics	-	I: aspart (premeal)	145	168 days	Yes
				C: RHI (premeal)	64		
Kawamori 2009	Japan	Multicentre, outpatients	2004-2006	I: glulisine (premeal)	132	28 weeks	Yes
				C: lispro (premeal)	135		
Provenzano 2001	Italy	-	-	I: lispro (premeal)	6	6 months ^b	Yes
				C: RHI (premeal)	6		
Raskin 2000	United States, Canada	Multicentre, outpatients	1997	I: aspart (premeal)	597	6 months	Yes
				C: RHI (premeal)	287		
Recasens 2003	Spain	-	-	I: lispro (premeal)	22	12 months	Yes
				C: RHI (premeal)	23		

(Ultra-)short-acting insulin analogues for adults with type 1 diabetes mellitus on multiple daily injections: a network meta-analysis (Review)

Table 1. Overview of studies included in synthesis (study characteristics, intervention and comparisons) (Continued)

Z011 2007	North America, Europe, South Africa	Multicentre, outpa- tients	1992-1993	I: lispro (premeal)	81	12 months	Yes
				C: RHI (premeal)	86		
Z013 2007	North America, Europe, South Africa, Australia	Multicentre, outpa- tients	1992-1993	I: lispro (premeal)	81	12 months	Yes
				C: RHI (premeal)	88		
Z015 2007	North America, Europe	Multicentre, outpa- tients	1993-1994	I: lispro (premeal)	50	12 months	Yes
				C: RHI (premeal)	48		
Ultra-short-acting insulins (4 studies)							
PRON- TO-T1DM ^c	United States, Japan, Ger- many	Multicentre, outpa- tients	2019-2020	I1: URLi (mealtime)	451	52 weeks	Yes
				C: lispro (mealtime)	442		
ONSET 1 ^d	Belgium, Canada, Czech Republic, Finland, Ger- many, Hungary, Poland, UK, United States	Multicentre, outpa- tients	2013-2015	I: faster aspart (mealtime)	381	52 weeks	Yes
				C: aspart (mealtime)	380		
ONSET 8 ^e	Austria, Bulgaria, Cana- da, Germany, India, Israel, Italy, Japan, Russian Fed- eration, Serbia, Taiwan, United States	Multicentre, outpa- tients	2018	I: faster aspart (mealtime)	342	26 weeks	Yes
				C: aspart (mealtime)	342		
Ma 2024	China, Argentina, Mexico	Multicentre, outpa- tients	29 May 2019-10 Jan 2022	I: URLi (mealtime)	176	26 weeks	Yes
				C: insulin lispro (mealtime)	178		

^aCross-over study and 24 weeks' intervention for each arm.

^bCross-over study with a total duration of one year; insulin was changed after six months (the six-month-long blocks were divided into two x three-month blocks with a normal and Mediterranean diet).

^{c,d,e}Insulin analogues injected post-meal were not included due to the author's exclusion (c,d) or the non-blinded arm (d).

"-" denotes not reported.

C: comparator; I: intervention; I1: intervention 1; I2: intervention 2; RHI: regular human insulin; SD: standard deviation; URLi: ultra-rapid lispro.

Table 2. Overview of studies included in synthesis (populations)

Study ID	Interven- tions(s) and compara- tor(s)	Age (mean/ range; years (SD)/ median (IQR))	Sex (% female)	HbA1c (mean; % (SD))	BMI (mean, kg/ m ² (SD))	Dura- tion of di- abetes (mean; years)	Co-medication (% of participants)	Co-morbidities (% of partici- pants)	Race/ethnicity (%)
Short-acting insulins									
Dreyer 2005	I: glulisine (premeal)	39.1 (12.1)	41.9	7.60 (0.963)	24.9 (3.4)	17.4 (10.9)	Glargine: once daily	-	-
	C: lispro (premeal)	37.9 (12.4)	42.9	7.58 (0.890)	25.1 (3.5)	15.6 (10.3)		-	-
Ferguson 2001	I: lispro (pre- meal)	46 (11)/19 to 65	45	9.0 (1.1)	25 (3)	26 (10)/10 to 45	Twice daily free- mixed insulin: 11 (33%)	-	-
	C: RHI (pre- meal)						Multiple injection regimen: 22 (67%)		
Home 2000	I: aspart (premeal)	38 (11)/ -	45	8.0 (1.2)	25.1 (3.1)	15 (10)	NPH insulin, once or twice daily	-	White: 99
	C: RHI (pre- meal)	38 (12)/ -	44	8.0 (1.2)	24.9 (3.0)	15 (10)		-	White: 99
Iwamoto 2001	I: aspart (premeal)	34 (16)/ -	59	7.5 (1.1)	22 (3)	11 (7)	Intermediate or long- acting insulin: once a day 131 (91.6%), twice a day 12 (8.4%)	-	Asian: 100
	C: RHI (pre- meal)	32 (13)/ -	66	7.6 (1.1)	22 (2)	11 (6)	Intermediate or long- acting insulin: once a day 56 (90.3%), twice a day 6 (9.7%)	-	Asian: 100
Kawamori 2009	I: glulisine (premeal)	38.9 (14.3)	82 (62.1)	7.44 (0.93)	23.11 (2.91)	12.8 (9.5)	Glargine (78.8 and 77.0% in the glulisine and lispro groups re- spectively) with the remaining patients receiving NPH in-		Asian: 100
	C: lispro (premeal)	38.8 (12.9)	72 (53.3)	7.50 (0.96)	22.81 (2.70)	11.1 (7.1)			Asian: 100

Table 2. Overview of studies included in synthesis (populations) *(Continued)*

							sulin; instructions for diet and exercise therapy		
Proven- zano 2001 ^d	I: lispro (pre- meal)	28 (-)/14 to 44	58	7.6 (0.5)	23 (3)	12 (3 to 20) ^c	Slow-acting insulin once at bedtime (100%)	-	-
	C: RHI (pre- meal)								
Raskin 2000	I: aspart (premeal)	39 (11)/ -	49	7.9 (1.1)	26 (4)	16 (10)	NPH insulin, once or twice daily (100%) ^b	-	White: 94
	C: RHI (pre- meal)	40 (12)/ -	47	8.0 (1.3)	26 (3)	16 (9)		-	White: 93
Recasens 2003	I: lispro (pre- meal)	24 (6)/ -	36	10.5 (2.4)	22 (1)	8.1 (3.8) weeks from diag- nosis	NPH insulin, once or twice daily (100%)	-	-
	C: RHI (pre- meal)	23 (5)/ -	39	11.4 (1.9)	21 (3)	8.1 (8.0) weeks from diag- nosis		-	
Z011 2007	I: lispro (pre- meal)	29 (-)/ -	49	8.2 (1.4)	24 (-)	12	Once or twice daily with ultralente ^a	-	-
	C: RHI (pre- meal)	32 (-)/ -	55	8.3 (1.7)	25 (-)	13		-	-
Z013 2007	I: lispro (pre- meal)	35 (-)/ -	49	8.3 (1.6)	24 (-)	13	Once or twice daily with NPH ^a	-	-
	C: RHI (pre- meal)	32 (-)/ -	48	8.1 (1.6)	24 (-)	11		-	-
Z015 2007	I: lispro (pre- meal)	24 (-)/ -	44	9.2 (2.2)	23 (-)	0.2	Once or twice daily with NPH or ultralen- te ^a	-	-
	C: RHI (pre- meal)	25 (-)/ -	33	8.8 (2.2)	23 (-)	0.2		-	-

Table 2. Overview of studies included in synthesis (populations) *(Continued)*

Ultra-short-acting insulins										
PRON-TO-T1DM	C: lispro (mealtime)	44.5 (13.6)	42.1	7.33 (0.67)	26.4 (4.3)	19.1 (12)	Glargine once daily: 49.5%	-	Race:	Ethnicity:
							Glargine twice daily: 6.8%		Asian: 17.6	Hispanic or Latino: 7.5
							Degludec: 43.7%		African American: 2	Not Hispanic or Latino: 89.8
									Multiple: 2.5	
									White: 77.8	Unknown or not reported: 2.7
	I1: URli (mealtime)	44.1 (13.7)	44.6	7.34 (0.65)	26.6 (4.2)	18.8 (12.3)	Glargine once daily: 47%		Race:	Ethnicity:
							Glargine twice daily: 9.3%		American Indian or Alaska: 0.2	Hispanic or Latino: 7.8
							Degludec: 43.7%		Asian: 19.1	Not Hispanic or Latino: 88.5
									African American: 1.6	
									Multiple: 2.2	Unknown or not reported: 3.8
									White: 76.7	
									Not reported: 0.2	
ONSET 1	I: faster aspart (mealtime)	46.1 (13.8)	44	7.6 (0.7)	26.4 (3.8)	20.9 (12.9)	Once or twice daily insulin detemir	Diabetic neuropathy: 22.3	Race:	Ethnicity:
								Diabetic retinopathy: 29.9	White: 95.3	Hispanic or Latino: 8.7
								Diabetic nephropathy: 8.9	Black or African American: 1.3	Not Hispanic or Latino: 91.3
									Asian: 1.3	
									Native Hawaiian or other	

Table 2. Overview of studies included in synthesis (populations) (Continued)

							Diabetic ketoacidosis: 2.1, Dyslipidaemia: 10.5 Hyperlipidaemia: 18.6 Hypertension: 38.3 Depression: 8.9 Coronary artery disease: 0.3		
							Pacific Islander: 0 American Indian or Alaska Native: 0 Other: 0 Not reported: 1.8		
							Diabetic neuropathy: 18.9	Race:	Ethnicity:
							Diabetic retinopathy: 28.7	White: 91.6	Hispanic or Latino: 4.2
							Diabetic nephropathy: 8.9	Black or African American: 2.4	Not Hispanic or Latino: 95.8
							Diabetic ketoacidosis: 1.6	Asian: 1.8	
							Dyslipidaemia: 8.2	Native Hawaiian or other Pacific Islander: 0.5	
							Hyperlipidaemia: 19.2	American Indian or Alaska Native: 0	
							Hypertension: 33.9	Other: 0.8	
							Depression: 5.0	Not reported: 2.9	
							Coronary artery disease: 0		
							Diabetic neuropathy: 31.9	Race:	Ethnicity:
								White: 64.0	
ONSET 8	I: faster aspart (mealtime)	41.48 (14.42)	46.2	7.46 (0.68)	25.06 (4.11)	17.58 (12.45)	Once daily degludec		

Table 2. Overview of studies included in synthesis (populations) *(Continued)*

C: aspart (mealtime)	40.77 (14.22)	47.7	7.41 (0.79)	25.06 (4.38)	16.74 (11.00)
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Diabetic retinopathy: 30.7	Asian: 33.9 Others (Black or African American, American Indian or Alaska Native): 2.1	Hispanic or Latino: 3.8
Diabetic nephropathy: 12.9		Not Hispanic or Latino: 96.2
Diabetic ketoacidosis: 0.6		
Dyslipidaemia: 12.6		
Hyperlipidaemia: 14.3		
Hypertension: 22.8		
Depression: 6.7		
Hepatic steatosis: 4.1		
Coronary artery disease: 0.6		
Diabetic neuropathy: 30.1	Race:	Ethnicity:
Diabetic retinopathy: 31.6	White: 57.9	Hispanic or Latino: 3.5
Diabetic nephropathy: 8.2	Asian: 40.1	
Diabetic ketoacidosis: 2.0	Others (Black or African American, American Indian or Alaska Native): 2.0	Not Hispanic or Latino: 96.5
Dyslipidaemia: 14.9, Hyperlipidaemia: 15.5		
Hypertension: 21.6		
Depression: 5.0		

Table 2. Overview of studies included in synthesis (populations) (Continued)

								Hepatic steato- sis: 4.1	
								Coronary artery disease: 1.5	
Ma 2024	I: URli (mealtime)	40.3 (13.4)	46.6	7.87 (0.95)	22.888 (3.144)	10.19 (9.13)	Metformin: 4.0% Glargine once daily: 91.5% Degludec once daily: 8.5%	-	Race: Asian: 81.8 Multiple: 2.8 White: 15.3
	C: lispro (mealtime)	43.6 (13.8)	43.8	7.82 (0.91)	26.662 (3.212)	10.49 (8.20)	Metformin: 5.1% Glargine once daily: 89.9% Degludec once daily: 10.1%		Race: Asian: 79.2 Multiple: 3.4 White: 17.4

^aAccording to IQWiG (2005), the original study reports mentioned 2 participants who took acarbose and 1 participant who took metformin during the study period. These participants were included in the per-protocol analysis.

^bThe list of co-medications was not included in the study report provided for the IQWiG report.

^c"12 (3-2)" in the publication. We assume this is a typing mistake, and it should be '12 (3-20)'.

^dEach arm also had 3 months' normal diet and 3 months' Mediterranean diet.

"-" denotes not reported.

BMI: body mass index; C: comparator; HbA1c: glycosylated haemoglobin A1c; I: intervention; IQR: interquartile range; RHI: regular human insulin; SD: standard deviation.